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A domain-specific cross-lingual semantic alignment learning model for low-resource languages

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ABSTRACT

Cross-lingual semantic alignment models facilitate the sharing and utilization of multilingual domain-specific data (e.g., medical, legal), offering cost-effective solutions for improving low-resource language tasks. However, existing methods are challenged by parallel data scarcity, semantic space heterogeneity, morphological complexity, and weak robustness-particularly for agglutinative languages. Therefore, this paper proposes CLWKD, a cross-lingual mapping and knowledge distillation framework. CLWKD leverages domain-specific pretrained models from high-resource languages as teachers and integrates multi-granularity alignment matrices with limited parallel data to guide cross-lingual knowledge transfer. CLWKD jointly learns multi-granularity semantic alignment mapping matrices at the token, word, and sentence levels from general-domain data. It eases domain data scarcity and helps bridge structural gaps caused by morphological and syntactic differences. To alleviate data sparsity and out-of-vocabulary issues in agglutinative languages, multilingual embedding sharing and morphological segmentation strategies are introduced. To improve the stability of unsupervised mapping training, generator pretraining is introduced and further combined with high-confidence word and sentence pairs to optimize the mapping matrix. To preserve alignment with fewer parameters, a parameter recycling and embedding bottleneck design is adopted. Experiments across the medical, legal, and educational domains on Mongolian-Chinese and Korean-Chinese language pairs demonstrate the effectiveness of CLWKD in three cross-lingual tasks.

1. Introduction

In recent years, pretrained language models (PLMs) have achieved remarkable progress in open-domain tasks for high-resource languages. However, owing to the severe scarcity of both general-domain and domain-specific corpora, low-resource languages still exhibit limited capability in domain-specific text mining, such as in the medical and legal fields. Cross-lingual semantic alignment models facilitate the sharing and utilization of domain data across multiple languages, providing an effective and low-cost approach to improve the performance of deep learning models for low-resource languages in specific domains. Therefore, exploring cross-lingual semantic alignment models tailored for low-resource languages has become increasingly important in domain-specific scenarios.

Current mainstream approaches to cross-lingual semantic alignment can be broadly categorized into three types: cross-lingual fine-tuning,

cross-lingual mapping, and cross-lingual knowledge distillation. These methods enhance the semantic alignment capability of multilingual models at the sentence level, word level, and intermodel level.

Cross-lingual fine-tuning methods aim to align sentence-level semantics by optimizing the [CLS] representation in multilingual PLMs. To further improve sentence alignment, researchers have introduced auxiliary token-level alignment tasks during training (Hu et al., 2020; Li et al., 2023; Zhang et al., 2023), and some studies leverage word alignment supervision to increase sentence-level performance (Artetxe et al., 2017b; Chousa et al., 2020; Lample et al., 2017; Lian et al., 2020; Miao et al., 2024; Schwenk et al., 2019). Recently, Zhu et al. (2024) argued that hard token-to-token alignment may disrupt sentence-level structure and thus proposed the use of alignment matrices to preserve contextual dependencies, coupled with a self-distillation mechanism to improve model robustness. To further improve multilingual embedding alignment for low-resource languages, Ai and Huang (2024) proposed

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initializing embeddings into similar geometrical structures across languages before training by modeling word frequency ranks from monolingual vocabularies. [Pham et al. \(2024\)](#) introduced an embedding initialization method that jointly leverages lexical and semantic alignment, along with a systematic search for the optimal vocabulary size. [Li and Li \(2024\)](#) proposed an angular optimization mechanism to preserve the structural relationships among sentence embeddings.

Cross-lingual mapping methods focus on word-level semantic alignment by learning a mapping matrix between word embeddings to project different languages into a shared representation space. This strategy is particularly well suited for low-resource languages ([Artetxe et al., 2018](#); [Mikolov et al., 2013](#); [Ruder et al., 2019](#); [Wang et al., 2019](#)). The approach was first introduced by [Mikolov et al. \(2013\)](#), and subsequent studies have incorporated adversarial training mechanisms to iteratively optimize the mapping matrix between a generator and a discriminator via parallel or contrastive corpora ([Bai et al., 2019](#); [Conneau et al., 2017](#); [Liang et al., 2023](#); [Tao et al., 2023](#); [Zhang et al., 2017](#)). Although such methods support weakly supervised or even unsupervised learning, the adversarial mechanism is often sensitive to initialization and prone to instability during training, especially with respect to agglutinative languages with complex morphological structures. Recently, [Hu and Xu \(2024\)](#) proposed aligning multiple subspaces to alleviate structural inconsistency in the embedding space, exhibited in improvements in medium-resource languages. This method struggles to address semantic misalignment caused by syntactic and morphological variations.

Cross-lingual knowledge distillation methods transfer semantic knowledge from high-resource languages to low-resource models via a teacher-student architecture. Typically, a high-resource PLM serves as the teacher to guide the student model in learning task-specific semantics and achieving cross-lingual representation alignment ([Ding et al., 2022](#); [Reimers & Gurevych, 2020](#); [Wang et al., 2023](#)). Unlike the previous two methods, knowledge distillation requires no large-scale labeled data in the target language and offers high transfer efficiency. It also enables the direct use of high-resource language representations in vertical domains to improve task performance for low-resource languages. In addition, the distillation mechanism naturally supports model compression, making it more suitable for deployment. Despite its theoretical advantages, the practical effectiveness of cross-lingual knowledge distillation remains constrained by two major challenges. First, the distillation process relies on high-quality domain-specific parallel corpora, which are extremely scarce for low-resource languages in specialized fields such as medicine and law. Second, the significant structural differences between the semantic spaces of the source and target languages often lead to unstable knowledge transfer. To alleviate the problem of semantic space heterogeneity, prior studies have explored techniques such as prototype modeling, similarity-based constraints, and contrastive learning ([Ge et al., 2023](#); [Park et al., 2024](#); [Yang et al., 2023](#)). For example, the IMASCL ([Park et al., 2024](#)) method addresses semantic space divergence by employing a soft contrastive learning strategy. Specifically, this method distills the similarity structure derived from a monolingual teacher model to guide the student model in aligning semantic representations across multiple languages. These methods perform well when applied to languages with similar syntactic and part-of-speech structures. Their alignment effectiveness remains limited when dealing with agglutinative languages that exhibit rich morphological variations and complex structures.

These challenges are particularly pronounced in China's multi-ethnic linguistic context. With the deepening of cultural integration, traditional Chinese medicine has evolved into a comprehensive system that includes Tibetan, Miao, Mongolian, and other ethnic medical traditions. Similarly, domains such as law and education also widely adopt minority languages. However, many of these languages—such as Mongolian, Tibetan, Kazakh, and Uyghur—suffer from severe data scarcity, strong agglutinative morphology, and flexible word order, which further complicate semantic modeling and cross-lingual knowledge transfer.

In summary, current cross-lingual semantic alignment for domain-specific tasks in low-resource agglutinative languages faces three core challenges: 1) Insufficient data coverage: The insufficient corpus coverage of low-resource languages (e.g., Mongolian) in multilingual pre-training, coupled with the scarcity of large-scale domain-parallel data, significantly hinders cross-lingual semantic alignment and domain knowledge transfer. 2) Semantic space heterogeneity: Significant differences in syntactic structures and lexical morphology between source and target languages lead to inconsistencies in semantic representation spaces, increasing the difficulty of alignment. 3) Strong agglutinative morphology: Agglutinative languages such as Mongolian exhibit complex word formation and rich inflection, which further exacerbates data sparsity and the out-of-vocabulary (OOV) problem.

Therefore, the objective of this study is to develop a cross-lingual semantic alignment model tailored for low-resource languages that can effectively address three key issues—domain-specific data scarcity, semantic space heterogeneity, and data sparsity caused by complex morphology—while ensuring model robustness.

To this end, we propose CLWKD, a cross-lingual mapping and knowledge distillation framework for multilevel semantic alignment in low-resource scenarios. The overall model architecture of CLWKD is illustrated in [Fig. 1](#). Compared with existing approaches, CLWKD not only introduces a multi-granularity collaborative strategy for semantic alignment, but also, for the first time, combines a general-domain mapping matrix with domain-specific knowledge distillation to address modeling challenges in extremely low-resource language settings. The key components are as follows:

First, to alleviate the problem of domain-specific data scarcity, we introduce a domain-specific pretrained model from a high-resource language as the teacher network. Using a general-domain mapping matrix and a small amount of domain parallel data as a bridge, we effectively transfer and align domain knowledge to a multilingual PLM. Unlike existing approaches, CLWKD reduces reliance on large-scale domain-specific data by leveraging a general-domain mapping matrix. It adaptively incorporates contextual representations through the mapping matrix to preserve lexical and syntactic structures from the general domain, thereby enhancing modeling capacity in low-resource scenarios. Compared with models such as IMASCL ([Park et al., 2024](#)), which focus on aligning the similarity structure distilled from teacher models, CLWKD places greater emphasis on maintaining multi-granularity consistency across lexical, syntactic, and semantic levels within the mapping space. Moreover, CLWKD further integrates domain knowledge distillation to achieve coordinated alignment of structural and semantic representations.

Second, to address the problem of semantic space heterogeneity, we jointly learn multi-granularity semantic alignment mapping matrices at the token, word, and sentence levels via general-domain parallel corpora. This design mitigates the substantial morphological and syntactic differences between the source and target languages. In contrast to prior methods based on single-granularity mappings (either at the word or sentence level), our approach employs a multi-granularity alignment strategy to leverage complementary information. Specifically, sentence-level alignment is guided by word-level supervision signals, and word-level alignment is supported by sentence-level contextual representations. We argue that in low-resource languages, word-level and sentence-level alignment are not isolated tasks but rather mutually reinforcing and coevolving processes. Relying solely on either layer may lead to semantic drift or grammatical inconsistency, especially for language pairs with significant structural divergence. Therefore, joint modeling across granularities offers a more robust and efficient solution. Compared with methods such as DM-BLI ([Hu & Xu, 2024](#)), which rely on bilingual dictionaries and are less sensitive to contextual information, CLWKD demonstrates stronger robustness in semantic alignment, particularly for structurally divergent and extremely low-resource language pairs.

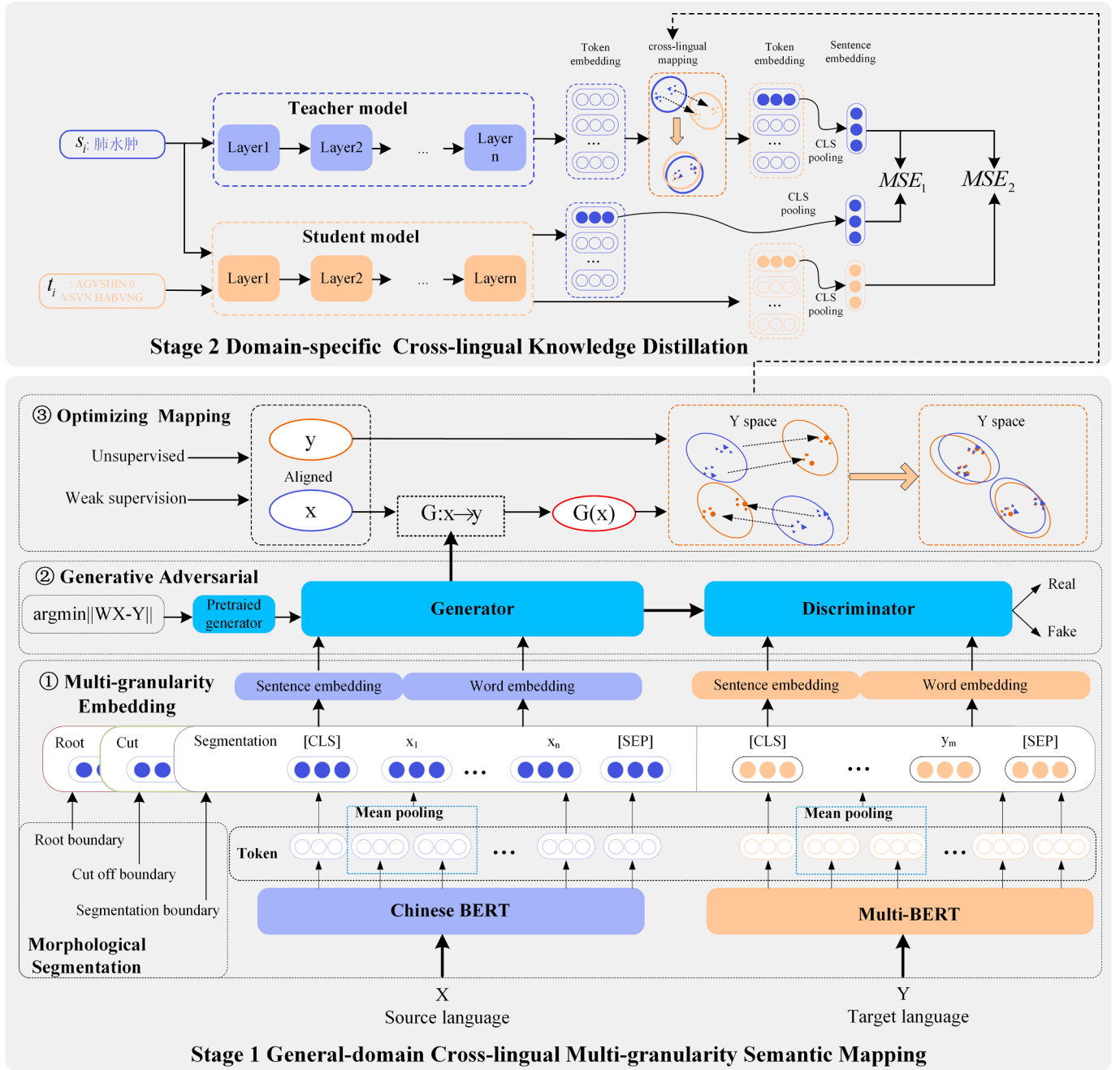


Fig. 1. The CLWKD framework consists of two stages. In the first stage: ① A morphological decomposition strategy is introduced to split words in the target language (e.g., Mongolian) into stems and affixes. This alleviates the issues of morphological complexity and OOV words, and constructs multi-granularity representations at the token, word, and sentence levels; ② A GAN structure is employed. The generator is pretrained to project source language embeddings into the semantic space of the target language across different granularities. The discriminator determines whether the projected embeddings resemble real target language representations, thereby guiding the generator to learn effective cross-lingual semantic mappings; ③ High-confidence word and sentence pairs are then leveraged to further optimize the mapping matrix under unsupervised or weakly supervised settings. In the second stage, a domain-specific PLM is used as the teacher model. With the multi-granularity mapping matrix learned in the first stage and a small amount of domain parallel data, knowledge distillation is performed to transfer and align domain knowledge from the high-resource language to the low-resource language model.

Third, to address the limited corpus coverage of low-resource languages (e.g., Mongolian) in multilingual pretraining, as well as the data sparsity and OOV issues caused by their complex agglutinative morphology, CLWKD proposes leveraging a shared multilingual PLM embedding layer combined with morphological decomposition strategies to enhance modeling effectiveness and cross-lingual alignment capability. Unlike existing approaches that rely on monolingual word frequency modeling (Ai & Huang, 2024) or optimize vocabularies through lexical

alignment (Pham et al., 2024) for cross-lingual space alignment, CLWKD enables Mongolian embeddings to directly benefit from the representation spaces of high-resource languages such as Chinese and English during the initialization stage, and further refines subword-level modeling through morphological segmentation to alleviate data sparsity issues. Notably, in real-world Mongolian domain corpora (e.g., medical texts), the frequent code-switching between Chinese and English provides a natural context for cross-lingual representation sharing, further

promoting semantic interaction and enhancing the model's effectiveness in practical tasks.

Finally, with respect to mapping matrix training, CLWKD first employs an adversarial network to generate an initial alignment structure, after which, it then iteratively improves the mapping performance through gradient-based optimization, guided by high-confidence word and sentence pairs. To mitigate instability and initialization sensitivity commonly observed in adversarial training, we introduce a generator pretraining mechanism, which enhances both the stability of the mapping matrix and its cross-lingual adaptability. For model compression, we adopt a combination of parameter recycling and an embedding bottleneck strategy to retain the semantic alignment capacity while reducing model size.

We conduct a systematic evaluation of CLWKD on two language pairs across three domains—medical, legal, and educational—covering three cross-lingual tasks. The results demonstrate that CLWKD has significant advantages in domain-specific word and sentence alignment tasks for low-resource agglutinative languages.

The main contributions of this paper are as follows:

We propose a unified framework, CLWKD, which integrates cross-lingual mapping and knowledge distillation. It is designed to address key challenges in domain-specific semantic alignment for low-resource languages—namely, the scarcity of domain parallel data, semantic space heterogeneity, and data sparsity caused by complex morphology. CLWKD effectively enhances word- and sentence-level semantic alignment across languages.

We design a context-aware alignment strategy based on general-domain corpora, which leverages a generative adversarial network (GAN) framework to jointly learn token-, word-, and sentence-level cross-lingual semantic mapping matrices. This is achieved through multiple techniques, including generator pretraining, gradient-based mapping optimization, shared multilingual PLM embeddings, and morphological decomposition.

We leverage multi-granularity mapping matrices and a small amount of domain-specific data as a bridge for cross-lingual knowledge distillation, enabling the transfer of high-resource domain knowledge to multilingual models for semantic alignment. During model compression, we employ parameter recycling and embedding bottleneck strategies to preserve alignment capacity.

In experiments conducted on two language pairs—Mongolian–Chinese and Korean–Chinese—across three cross-lingual tasks (word alignment, bitext retrieval, and entity recognition) and multiple domains (medical, legal, and educational), the results show that CLWKD significantly outperforms existing methods. Moreover, the compressed variant, CLWKD_S, also achieves strong performance and competitive effectiveness.

2. Related work

2.1. Pre-trained language models

Multilingual pre-trained models. To address the performance issues of PLMs for low-resource languages, researchers have proposed multilingual PLMs (Conneau, 2019; Liu, 2020; Shliazhko et al., 2022; Xue, 2020), which provide the possibility for learning representations of multiple languages jointly. Multilingual bidirectional encoder representations from transformers (mBERT) (Kenton & Toutanova, 2019) was pre-trained on Wikipedia data in 104 languages and achieved significant performance in downstream tasks. XLM (Conneau & Lample, 2019) introduced the translation language modeling objective, leveraging parallel data to enhance cross-lingual transferability. XLM-R (Conneau, 2019) continued training on RoBERTa (Liu, 2019) via parallel corpora.

Domain-specific pre-trained language models. Owing to the abundance of specialized medical terminology in medical texts, general PLMs often perform poorly in extracting information from medical content. In recent years, researchers have developed domain-specific

PLMs for the medical field. Lee et al. (2020) proposed BioBERT, a pre-trained model that is similar in structure to BERT and is further trained on biomedical abstracts from PubMed and full-text articles from PMC. Symeonidou (Symeonidou et al., 2019) introduced a transfer learning model for named entity recognition (NER) that is based on BioBERT-BiLSTM-CRF, which demonstrated improved performance over previous NER models across three medical datasets. Rasmy et al. (2021) developed Med-BERT, a pre-trained model trained from scratch using abstracts from PubMed and full text from PMC, which significantly boosts performance in small sample tasks. Yang (2021) proposed MedBERT, a model designed for Chinese clinical natural language processing tasks. Subsequently, other researchers further fine-tuned and improved MedBERT (Gan et al., 2021). Zhang et al. (2020) from Alibaba introduced MC-BERT, a pre-trained model for Chinese medical texts that is built on BERT and trained with Chinese medical data. ERNIE-Health, which is based on Baidu's ERNIE model, learns the relationships between symptoms and treatment plans through medical entity masking and question-answer matching tasks and outperformed BERT on the CCKS2019 medical entity recognition and question-answer matching tasks.

To improve semantic understanding and task performance in the legal domain, researchers have developed several Chinese legal pre-trained language models, among which Lawformer (Xiao et al., 2021) and LegalRoBERTa¹ are representative. The Lawformer is based on the Longformer architecture with sparse attention, enabling efficient processing of lengthy legal documents. Pretrained on large-scale real-world legal texts, it has been applied to tasks such as legal classification, statute matching, and question answering. LegalRoBERTa is a RoBERTa-based model further pretrained on Chinese legal corpora, retaining general language understanding while adapting to legal-specific tasks such as document classification and legal entity recognition.

In the education domain, TAL-EduBERT² and EduBERT (Clavié & Gal, 2019) are two notable Chinese pretrained models. TAL-EduBERT, developed by the TAL Education Group, is pretrained on large-scale educational data, including textbooks, teaching materials, and student answers. EduBERT, built on BERT, is further pretrained on education-related corpora and is used in tasks such as academic writing analysis and educational resource recommendation.

2.2. Cross-lingual fine-tuning and mapping semantic alignment

Currently, cross-lingual fine-tuning semantic alignment methods based on multilingual PLMs are primarily used for learning sentence-level semantic alignment, while cross-lingual mapping methods are mainly applied to word-level semantic alignment. The following section provides a detailed explanation.

Cross-lingual fine-tuning for sentence-level semantic alignment methods. InfoXLM (Sun et al., 2019) extends the XLM-R model, which is pre-trained on a large corpus of parallel data through masked language modeling and translation language modeling. XLM-Align (Chi et al., 2021), which also uses many parallel sentences, first performs self-supervised word alignment on these sentences and then trains by masked language modeling. Recently, the importance of token-level auxiliary tasks has been recognized. Zhang et al. (2023) achieved significant results on the XTREME benchmark (Hu et al., 2020) by using synonym-based token alignment. Li et al. (2023) improved sentence-level semantic alignment models on multilingual PLMs through token alignment tasks. Owing to the scarcity of parallel corpora, research on cross-lingual sentence embeddings for low-resource languages has lagged. To address this, researchers have proposed enhancing cross-lingual sentence embeddings for low-resource languages by utilizing word alignment tasks on top of multilingual PLMs (Miao et al., 2024). The basic idea of unsupervised sentence alignment is to further achieve

¹ <https://data.thunlp.org/legal/LegalRoBERTa.zip>

² <https://github.com/tal-tech/edu-bert>

sentence-level alignment on the basis of prealigned word embeddings (Artetxe et al., 2017b; Lample et al., 2017). Schwenk et al. (2019) mapped word embeddings from different languages to a unified vector space to obtain bilingual sentence pairs. Lian et al. (2020) proposed a similarity calculation method that maps monolingual corpora from each language to a low-dimensional semantic space via autoencoders. Then, the difference between each language's distribution in this semantic space and a central distribution is measured via the Wasserstein distance, and this distance is minimized to establish alignment relationships with the central distribution. Chousa et al. (2020) determined the optimal sentence alignment by predicting the similarity between segments in source and target language sentences.

Cross-lingual mapping for word-level semantic alignment methods. Mikolov et al. (2013) trained word vectors for two languages separately and then learned a linear mapping matrix between vector spaces from a small bilingual dictionary. Building on this, Xing et al. (2015) proposed normalizing word vectors so that their inner product and cosine distance are equal. Artetxe et al. (2017a) used self-learning methods to iteratively optimize the mapping matrix. With the development of pre-trained model techniques, dynamic word embedding models have emerged (Kenton & Toutanova, 2019; Liu et al., 2024). Wang et al. (2019) learned cross-lingual mapping matrices for dynamic word embeddings from parallel sentences and demonstrated that their performance exceeds that of static word embedding mapping matrices. Li et al. (2022) proposed a two-stage contrastive learning framework. In the first stage, contrastive learning is used to refine cross-lingual linear mappings between static word embeddings; in the second stage, contrastive fine-tuning of multilingual BERT enhances word-level translation capabilities. Compared with supervised learning methods, unsupervised and weakly supervised methods are more suitable for scenarios where aligned data are scarce in low-resource languages. Zhang et al. (2017) introduced GANs and used adversarial training to match distributions for bilingual word alignment. Conneau et al. (2017) generated high-quality aligned dictionaries after GAN training to optimize transformation matrices and proposed the cross-domain similarity local scaling (CSLS) metric to address the hubness problem in high-dimensional spaces by calculating the similarity between cross-lingual vocabularies. Bai et al. (2019) learned a shared semantic space for the source and target languages and used this space for unsupervised bilingual dictionary extraction. Tao et al. (2023) proposed an unsupervised word alignment method based on evolutionary algorithms and bidirectional recurrent GANs to address the instability of GAN training. Liang et al. (2023) developed a framework for constructing multilingual embeddings for domain transfer learning by extracting pivot words from current data and incorporating them as weight factors into embedding alignment. Recently, Recently, Hu and Xu (2024) further proposed a multi-subspace alignment strategy to enhance the robustness of embedding alignment.

In contrast to these approaches, this paper combines multiple strategies to jointly fine-tune and learn multi-granularity cross-lingual semantic mapping matrices for token, word, and sentence embeddings between Mongolian and Chinese. It fully leverages the interdependencies between word alignment and sentence alignment tasks while effectively

addressing challenges related to syntactic structure and morphological differences.

2.3. Cross-lingual knowledge distillation for knowledge transfer and semantic alignment

As mentioned earlier, knowledge distillation has three major advantages: model compression, knowledge transfer, and semantic alignment.

In the context of model compression, BERT-PKD (Sun et al., 2019) extracts knowledge from both the final layer and intermediate layers of the model during fine-tuning. DistilBERT (Sanh, 2019) performs distillation during pre-training, halving the number of hidden layers in BERT. TinyBERT (Jiao et al., 2019) extracts knowledge from BERT during both pre-training and fine-tuning. Mobile-BERT (Sun et al., 2020) refines BERT into a model with smaller dimensions for each layer. Additionally, Wang et al. (2020) conducted deep self-attention distillation of BERT.

In the areas of cross-lingual knowledge transfer and vector space alignment, researchers have recently made significant progress. Reimers and Gurevych (2020) proposed a multilingual knowledge distillation method that transfers source language vector spaces to various target language spaces through parallel corpora, achieving vector space alignment. To obtain compact models with cross-lingual matching capabilities, Ding et al. (2022) proposed a multistage distillation framework that combines contrastive learning, bottleneck, and parameter cycling strategies to align multilingual vector spaces while transferring knowledge from large models to smaller models. Wang et al. (2023) used cross-lingual knowledge distillation to transfer temporal knowledge from high-resource language knowledge graphs to low-resource language knowledge graphs, addressing the issue of limited cross-lingual alignment corpora between two temporal knowledge graphs by generating pseudoalignments via time information extracted by the representation module. Yang et al. (2023) proposed a bidirectional semantic alignment-based similarity measurement method for mapping representations of different languages into the same space, addressing issues of differing representation spaces between teacher and student models. Ge et al. (2023) addressed the problem of feature distribution differences between source and target languages in teacher-student networks by adjusting the distance between source and target language features via prototype networks for feature alignment.

Unlike these knowledge distillation methods, CLWKD maps source language domain vectors to multilingual target language domain spaces via a universal cross-lingual matrix W , achieving domain data augmentation and addressing the issue of differing language spaces. During model compression, CLWKD initializes the compressed model with the transformer layers of CLWKD and uses bottleneck layers to reduce the embedding size.

2.4. Mongolian

Mongolian, like Japanese, Korean, and Turkish, is an agglutinative language. Fig. 2 illustrates the structure of the Mongolian word “factory”, which consists of a root (or stem) connected to a varying number of suffixes. The root forms the basis for deriving new words and is the

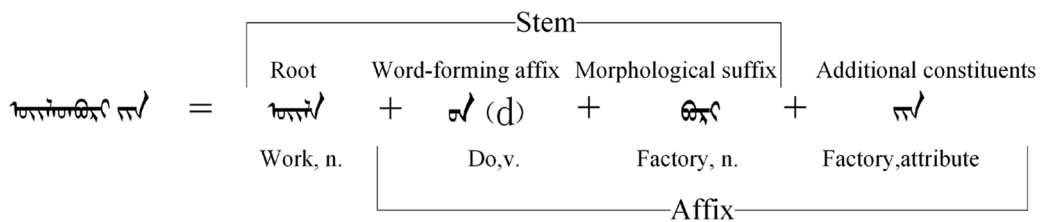


Fig. 2. Example of the morphological structure of the Mongolian word “factory.” The root conveys basic meaning, such as “work” (noun). This is followed by the sequential addition of word-forming affixes that change the part of speech: first deriving a verb (e.g., “do”) and then further deriving a noun (e.g., “factory”). In addition, additional constituents may be attached to modify or constrain the noun, such as in the “factory attribute”.

Mongolian	English translation:
ᠪᠢᠳᠡ ᠬᠠᠮᠲᠤᠨ ᠡᠳᠤᠢ ᠪᠠᠭᠤᠰᠠᠨ ᠤᠨ ᠭᠠᠶᠢᠬᠢᠮᠠᠷ	It's amazing how long we've been together
	Latin transliteration:
	BIDE HAMTV-DAGAN EDUI VDAGSAN-I GAYIHAMAR
	Segmentation:
	BIDE HAMTV -DAGAN EDUI VDAGSAN -I GAYIHAMAR
	Cut:
	BIDE HAMTV EDUI VDAGSAN GAYIHAMAR
	Root:
	BIDE HAM EDUI VDA GAYIHA

Fig. 3. Example of latin transliteration of Mongolian text. It includes three processing strategies for transliteration: Segmentation of affixes—separating affix components from the original Mongolian word; Affix removal—removing affixes and retaining only the core part of the word; Root retention—preserving only the root to retain the basic semantic meaning of the word.

fundamental unit preserving the word's meaning. Suffixes attached to the root alter the word's part of speech and meaning, significantly impacting the semantic context of sentences. The root combined with several suffixes forms a stem. Suffixes that are attached to the root (or stem) through a narrow, uninterrupted space are called affixes, which indicate grammatical meanings such as case, number, and possession. A single root (or stem) can have multiple affixes to express different grammatical meanings (Wang, 2009). Owing to the complex morphological changes in Mongolian word formation, various roots (or stems) with different suffixes can produce approximately one million Mongolian vocabulary items. Examples of Mongolian Latin transliteration, where affixes are segmented and removed and the stem is retained, are shown in Fig. 3. Studies have shown that segmenting Mongolian text can alleviate data sparsity issues (Hou et al., 2009; Li et al., 2017; Ren et al., 2019).

3. Method

The CLWKD framework consists of two stages, as illustrated in Fig. 1.

In the first stage (Section 3.1), we construct a unified cross-lingual multi-granularity semantic space based on general-domain parallel data between the source and target languages. This stage consists of three functional modules: In module ①, we apply a morphological decomposition strategy to the target language (e.g., Mongolian). Each word is segmented into a stem and its affixes to address the challenges of morphological complexity and out-of-vocabulary issues commonly found in agglutinative languages. Based on a pretrained language model, we construct shared token, word, and sentence-level embeddings using a unified embedding layer. In Module ②, we introduce a generative adversarial network. The pretrained generator projects source-language embeddings into the target-language semantic space. The discriminator then determines whether these projected embeddings come from real target-language representations. This adversarial setup helps the generator learn an effective cross-lingual semantic mapping. In module ③, we incorporate high-confidence word and sentence pairs to refine the mapping matrix under unsupervised or weakly supervised conditions. This step improves both the accuracy and stability of the learned alignment.

In the second stage (Section 3.2), a domain-specific PLM trained on a high-resource language serves as the teacher model. The multi-granularity mapping matrices learned in Stage 1, along with a small amount of domain-specific parallel data, are used as a bridge to transfer and align domain knowledge to the multilingual PLM for the low-resource language. During this process, we apply a mean squared error (MSE) loss to align the [CLS] representations of the source and target languages at the sentence level. Meanwhile, the student model incorporates a bottleneck embedding structure and a parameter recycling mechanism to enable model compression while preserving cross-lingual alignment capability.

The following sections provide detailed descriptions of each module and its implementation.

3.1. Cross-lingual multi-granularity semantic mapping

In cross-lingual domain semantic alignment tasks involving low-resource agglutinative languages, the scarcity of domain-specific parallel data and the structural differences between languages lead to semantic space inconsistencies, further increasing the difficulty of cross-lingual alignment. To address these challenges, by leveraging multi-granularity mapping matrices from general-domain data, the model first learns morphological and syntactic information across different granularities and languages, which reduces the divergence between semantic spaces of different languages. In addition, word-level and sentence-level alignment tasks can provide mutual supervision, enabling better overall alignment. After learning interlingual features from general-domain data, the model can then focus on the alignment and transfer of domain-specific knowledge during the knowledge distillation stage, thereby reducing reliance on domain-specific data.

Therefore, our first objective is to map the source language's token, word, and sentence embeddings to the corresponding representations in the target language. Given source language vectors $X = x_1, x_2, x_3, \dots, x_{|X|}$ and target language vectors $Y = y_1, y_2, y_3, \dots, y_{|Y|}$, we aim to learn a linear mapping matrix W between the source and target spaces so that WX is mapped as close as possible to Y .

$$W^* = \arg \min_{W \in \mathbb{M}_d(\mathbb{R})} \|WX - Y\| \quad (1)$$

where d is the embedding dimension; $\mathbb{M}_d(\mathbb{R})$ is the real space of matrices of size $d \times d$; and X and Y are alignment matrices of size $d \times |X|$ that represent token, word, and sentence embeddings in the source and target languages.

3.1.1. Multi-granularity embedding representation

Token embedding learning. Some existing studies (Ai & Huang, 2024), such as those based on monolingual word frequency modeling, perform coarse-grained spatial structure alignment across multiple languages by ranking word frequencies. However, these approaches overlook true cross-lingual interaction information and do not optimize the morphological characteristics of agglutinative languages, resulting in clear limitations when modeling low-resource domain corpora, particularly for morphologically complex languages such as Mongolian. To address this issue, we further analyze the characteristics of low-resource languages (e.g., Mongolian) within multilingual PLMs. Due to limited corpus coverage, Mongolian representations in PLMs often suffer from data sparsity and OOV problems. In addition, in real-world domain corpora (such as Mongolian medical texts), we observe frequent code-switching between Mongolian, Chinese, and English terms, thus providing a natural semantic bridge for cross-lingual representation sharing. This phenomenon helps the model to better capture cross-lingual associations and enhances performance in practical tasks. To fully exploit this characteristic, we first perform Latin transliteration for Mongolian, mapping its character representation into the vocabulary space supported by multilingual PLMs. This strategy maximizes the reuse of existing subword fragments, improves representation consistency between Mongolian and other languages, and effectively mitigates the OOV problem. On this basis, we further initialize Mongolian token embeddings using the multilingual PLM's embedding layer, enabling Mongolian representations to directly inherit semantic knowledge from high-resource languages such as Chinese and English at the early stage of training.

We then input the source language X into its domain-specific PLM (e.g., Chinese-BERT) and the target language Y into a multilingual PLM (e.g., mBERT) and perform contextual encoding for their respective token representations. Specifically, let the input be a sentence-level token sequence consisting of n vectors $x_1, x_2, \dots, x_n$, and the output be a new set of vectors $o_1, o_2, \dots, o_n$ corresponding to each input vector. Different three-parameter matrices W^q, W^k, W^v are used to map x_i into

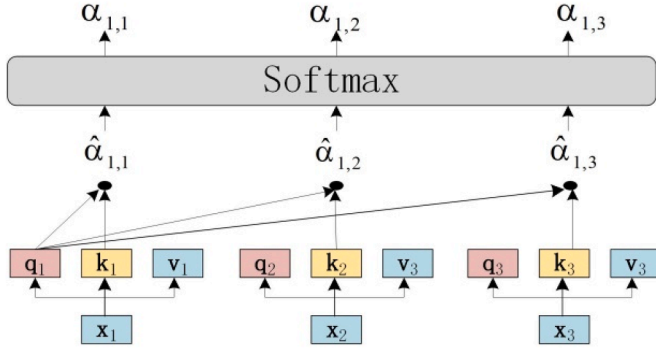


Fig. 4. Illustration of self-attention weight computation.

$q_i = W^q x_i$, $k_i = W^k x_i$, $v_i = W^v x_i$, which denote the vectors corresponding to the query, key and value, respectively. The formula for o_i is as follows:

$$o_i = \sum_{j=1}^n \alpha_{i,j} v_j, \quad \alpha_{i,j} = \text{softmax}(\hat{\alpha}_{i,j}), \quad \hat{\alpha}_{i,j} = \text{attn}(q_i, k_j) \quad (2)$$

where $\alpha_{i,j}$ denotes the attentional weight between x_i and x_j , with larger values indicating greater correlation, as illustrated in Fig. 4.

When self-attention is normalized, it is not possible to assign larger attention values to multiple inputs at the same time, so multiple sets of self-attention models are used to produce multiple sets of different attention results, thus enhancing the expressive power of the model, as shown in Eq. (3):

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (3)$$

where each attention head is computed as $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$, and Q , K , and V are the input token embedding matrices; and W^O denotes the output projection matrix.

Word embedding learning. In our experiments, we selected Chinese, Mongolian, Korean, and English as target languages and applied language-specific preprocessing accordingly. For Chinese texts, we used the Jieba tokenizer for word segmentation. In contrast, Mongolian, Korean, and English naturally use whitespace to separate words and thus do not require additional segmentation. However, when PLM tokenizers are used, words in these languages may still be further split into multiple subword tokens, which can hinder accurate word-level semantic modeling. To address this issue, we introduce word boundary information to identify the token span corresponding to each original word and compute word-level representations via mean pooling. Specifically, we let the k -th word w_k span the token sequence positions (i_k, j_k) . The word-level embedding of w_k is computed as follows:

$$w_k = \frac{1}{j_k - i_k + 1} \sum_{m=i_k}^{j_k} t_m \quad (4)$$

Here, $t_m \in \mathbb{R}^d$ denotes the embedding of the m -th token, and d is the embedding dimension. The denominator $j_k - i_k + 1$ represents the number of tokens corresponding to the word, thereby serving as a normalization factor to ensure balanced representation across words of varying lengths.

To address the data sparsity caused by rich morphological variations in agglutinative languages (e.g., Mongolian), we adopt a morphological decomposition strategy. Each word w_i is decomposed into a sequence of tokens $\{x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(k)}\}$. The decomposition granularity is defined as follows:

- $\text{root}(w_i) = \{x_i^{(1)}, \dots, x_i^{(r)}\}$: the root of the word;
- $\text{stem}(w_i) = \{x_i^{(1)}, \dots, x_i^{(m)}\}$: the stem of the word, where $m \geq r$;

- $\text{additional constituents}(w_i) = \{x_i^{(m+1)}, \dots, x_i^{(k)}\}$: the additional constituents.

We design three decomposition strategies for constructing subword-level word embeddings using mean pooling:

Root Strategy. Only the root is retained:

$$e_{\text{Root}}(w_i) = \frac{1}{r} \sum_{j=1}^r \text{Emb}(x_i^{(j)}) \quad (5)$$

Seg Strategy. The stem and affix are separated into two components:

$$e_{\text{Seg}}^{\text{stem}}(w_i) = \frac{1}{m} \sum_{j=1}^m \text{Emb}(x_i^{(j)}) \quad (6)$$

$$e_{\text{Seg}}^{\text{affix}}(w_i) = \frac{1}{k-m} \sum_{j=m+1}^k \text{Emb}(x_i^{(j)})$$

Cut Strategy. The additional constituents are removed, and only the final stem is preserved:

$$e_{\text{Cut}}(w_i) = \frac{1}{m} \sum_{j=1}^m \text{Emb}(x_i^{(j)}) \quad (7)$$

Here, $\text{Emb}(\cdot)$ denotes the embedding of each subword, and the mean operator is used to aggregate multiple subword embeddings into a unified word-level representation.

Through this mechanism, the model can effectively unify the representation of words with complex morphology, alleviating the issue of word-level distribution sparsity. A schematic of the decomposition process is shown in Fig. 2, and the effects of this decomposition are discussed in Section 5.1.

Sentence embedding learning. At the sentence level, to obtain the global semantic representation, we use the representation vector at the [CLS] position as the sentence embedding:

$$s = x^{[\text{CLS}]} \in \mathbb{R}^d \quad (8)$$

Here, $x^{[\text{CLS}]}$ is the output of the [CLS] token at the beginning of the sentence in the PLM, representing the aggregated semantic representation of the entire sentence.

3.1.2. Generative adversarial layer

To achieve semantic alignment between the source and target languages under unsupervised or weakly supervised conditions, we introduce a generative adversarial network structure. This structure consists of a generator and a discriminator, with the goal of learning a cross-lingual mapping matrix W such that the source language embeddings approximate the distribution of the target language.

The core of the generator is a trainable mapping matrix $W \in \mathbb{R}^{d \times d}$, which projects the source language embedding x_i into the target language embedding space, such that $\hat{y}_i = W x_i$. Ideally, the distribution of $\hat{y}_i$ should match the distribution of the real target language embedding y_i as closely as possible. The goal of the generator is to deceive the discriminator into believing that the mapped representations come from the target language. The loss function for the generator is as follows:

$$L_G(W | \theta_D) = -\frac{1}{n} \sum_{i=1}^n \log P_{\theta_D}(\text{source} = 0 | W x_i) - \frac{1}{m} \sum_{i=1}^m \log P_{\theta_D}(\text{source} = 1 | y_i) \quad (9)$$

The discriminator D is a binary classification network. The training objective of the discriminator is to classify real target language embeddings y_i as positive, and to classify the projected embeddings from the generator $\hat{y}_i = W x_i$ as negative. Accordingly, the binary cross-entropy loss of the discriminator is defined as:

$$L_D(\theta_D | W) = -\frac{1}{n} \sum_{i=1}^n \log P_{\theta_D}(\text{source} = 1 | W x_i)$$

$$-\frac{1}{m} \sum_{i=1}^m \log P_{\theta_D}(\text{source} = 0 \mid y_i) \quad (10)$$

To enhance robustness, we apply label smoothing to the positive samples. Let α denote the smoothing factor, then the label for target language samples is set to $1 - \alpha$:

$$\begin{aligned} L_D(\theta_D \mid W) = & -\frac{1}{n} \sum_{i=1}^n \log P_{\theta_D}(\text{source} = 1 \mid W x_i) \\ & -\frac{1}{m} \sum_{i=1}^m \left[(1 - \alpha) \log P_{\theta_D}(\text{source} = 0 \mid y_i) \right. \\ & \left. + \alpha \log P_{\theta_D}(\text{source} = 1 \mid y_i) \right] \end{aligned} \quad (11)$$

Algorithm 1 Adversarial training loop.

Require: Number of epochs N , batch size B , dataset D , BERT encoder

$\mathcal{M}$, mapping module M , discriminator D , statistics dictionary stats

Ensure: Optimized mapping M^*

```

1: for  $n_{\text{epoch}} = 1$  to  $N$  do
2:    $n_{\text{dis}} \leftarrow 0$ ,  $n_{\text{map}} \leftarrow 0$ ,  $n_{\text{proc}} \leftarrow 0$ 
3:   for each batch  $(X_a, M_a, X_b, M_b, B_a, B_b, B_a^{\text{mask}}, B_b^{\text{mask}}, \text{idx}) \in D$  do
4:      $H_a \leftarrow \mathcal{M}(X_a, M_a)$ ,  $H_b \leftarrow \mathcal{M}(X_b, M_b)$ 
5:      $W_a \leftarrow \text{token2word}(H_a, B_a, B_a^{\text{mask}})$ 
6:      $W_b \leftarrow \text{token2word}(H_b, B_b, B_b^{\text{mask}})$ 
7:     try:
8:        $\text{dis\_step}(W_a, W_b, \text{stats})$ 
9:     catch:
10:      Log idx
11:
12:     $n_{\text{dis}} \leftarrow n_{\text{dis}} + 1$ 
13:    if  $n_{\text{dis}} \bmod \text{dis\_steps} = 0$  then
14:      try:
15:         $\text{mapping\_step}(\text{stats})$ 
16:      catch:
17:        Log idx
18:
19:     $n_{\text{map}} \leftarrow n_{\text{map}} + 1$ 
20:  end if
21:  if  $n_{\text{dis}} \bmod \text{save\_every} = 0$  then
22:     $\text{metric}_{\text{sent}} \leftarrow \text{eval\_sim}()$ 
23:     $\text{metric}_{\text{word}} \leftarrow \text{dist\_mean\_cosine}()$ 
24:     $\text{score} \leftarrow \lambda \cdot \text{metric}_{\text{sent}} + (1 - \lambda) \cdot \text{metric}_{\text{word}}$ 
25:     $\text{acc}_D \leftarrow \text{eval\_dev\_dis}()$ 
26:    if  $\text{score} > \text{best\_score}$  then
27:      Save model checkpoint
28:    end if
29:  end if
30: end for
31: Save model: "epoch- $n_{\text{epoch}}$ "
32: Update learning rate using  $\text{metric}_{\text{word}}$ 
33: if learning rate  $< \text{min\_lr}$  then
34:   break
35: end if
36: end for

```

In our experiments, the discriminator is implemented as a multi-layer perceptron (MLP). Let the input be an embedding vector $\mathbf{z} \in \mathbb{R}^d$, with L hidden layers and hidden dimension h . The structure of the discriminator is defined as follows:

$$\begin{aligned} \mathbf{h}^{(0)} &= \text{Dropout}_{\text{in}}(\mathbf{z}) \\ \mathbf{h}^{(l)} &= \text{Dropout}(\text{LeakyReLU}(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)})), \quad \text{for } l = 1, \dots, L \quad (12) \\ D(\mathbf{z}) &= \sigma(\mathbf{W}^{(L+1)} \mathbf{h}^{(L)} + \mathbf{b}^{(L+1)}) \end{aligned}$$

Here, $\mathbf{W}^{(1)} \in \mathbb{R}^{h \times d}$ denotes the weight matrix of the first hidden layer; $\mathbf{W}^{(l)} \in \mathbb{R}^{h \times h}$ represents the weight matrices of the remaining hidden lay-

ers for $l = 2, \dots, L$; and $\mathbf{W}^{(L+1)} \in \mathbb{R}^{1 \times h}$ denotes the weight matrix of the output layer. The activation function is defined as:

$$\text{LeakyReLU}(x) = \max(0.2x, x) \quad (13)$$

The output is passed through a sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (14)$$

It is well known that although GANs can be applied for unsupervised alignment learning, their training process is highly challenging. In particular, during the early stages of training, there is often a capability imbalance between the generator and the discriminator. The discriminator typically converges faster and dominates the adversarial process. This imbalance hinders the optimization of the generator and leads to poor alignment quality.

To address this issue, we introduce a pretraining phase, where a small set of high-quality word pairs is used to train an initial mapping matrix W_0^* . This initialization enhances the generator's alignment capability in the early stages, accelerates model convergence, and improves the semantic consistency of the generated samples. Specifically, we obtain a set of word pairs $\{(x_i, y_i)\}_{i=1}^N$ from 5000 high-quality parallel sentence pairs via fast-align (Dyer et al., 2010). Then, using the obtained word pairs, we minimize the following objective:

$$W_0^* = \arg \min_W \sum_{i=1}^N \|W x_i - y_i\|_2^2 \quad (15)$$

This problem has an analytical solution, and the optimal solution is obtained through singular value decomposition (SVD):

$$W_0^* = UV^\top, \quad \text{where } U\Sigma V^\top = \text{SVD}(YX^\top) \quad (16)$$

By fully exploring the interdependencies between word alignment and sentence alignment tasks, this paper introduces two cross-lingual similarity evaluation strategies during training, which are used for model parameter selection and alignment performance evaluation: word mean similarity (Conneau et al., 2017) and sentence similarity.

Word Mean Similarity. This metric measures the closeness between the source and target language embedding spaces. Let $\{x_i\}_{i=1}^{1000}$ represent the embeddings of the 1000 most frequent words in the source language. For each word x_i , we retrieve the most similar word $\hat{y}_i$ from the target language embedding set and compute their cosine similarity:

$$\text{Words_Mean} = \frac{1}{1000} \sum_{i=1}^{1000} \cos(x_i, \hat{y}_i) \quad (17)$$

Sentence Similarity. This metric is calculated via the [CLS] token representation as the global semantic embedding of the sentence. The cosine similarity between the source language sentence s_x and the target language sentence s_y is computed as:

$$\text{Sent}_{\text{sim}} = \cos(s_x, s_y) = \frac{\mathbf{s}_x^\top \mathbf{s}_y}{\|\mathbf{s}_x\| \cdot \|\mathbf{s}_y\|} \quad (18)$$

We define a composite similarity metric that integrates both word-level and sentence-level information:

$$\text{Score}^{(t)} = \lambda \cdot \text{Sent}_{\text{sim}}^{(t)} + (1 - \lambda) \cdot \text{Words_Mean}^{(t)} \quad (19)$$

The entire adversarial training process consists of the following steps (see Algorithm 1):

- 1) Initialize the mapping matrix W and discriminator parameters θ_D ;
 - 2) Fix W and update the discriminator D , minimizing $\mathcal{L}_D$;
 - 3) Fix the discriminator and update the generator W , minimizing $\mathcal{L}_G$;
- Repeat Steps 2)-3) until the discriminator can no longer distinguish between real and generated samples (training converges).

3.1.3. Optimizing the mapping layer

After GAN training, the mapping matrix is further optimized by constructing high-quality aligned lexicons. We use both unsupervised and weakly supervised methods to build these aligned lexicons.

Unsupervised method: Using the mapping matrix to project bilingual word embeddings into a shared space, aligned lexicons are constructed by calculating the word similarity with CCLS:

$$\text{CCLS}(Wx_s, y_t) = 2 \cos(Wx_s, y_t) - r_T(Wx_s) - r_S(y_t) \quad (20)$$

where $r_T(Wx_s)$ is the average cosine similarity of the top k most similar word vectors in the target vector space after the source word vector x is mapped by the generator, and $r_S(y_t)$ is the average cosine similarity of the top k most similar word vectors in the source word vector space for the target vector y .

$$W(G(x), y_1) = \begin{cases} \text{Aligned} & y_1 - y_2 \geq \mu \\ \text{Unaligned} & y_1 - y_2 < \mu \end{cases} \quad (21)$$

where y_i represents the top k most similar words in the target language space found for the source language word x . When $y_1 - y_2$ is greater than the threshold, the corresponding target language-aligned word is considered found; otherwise, it is considered not found.

Weakly supervised method: Unsupervised contextual word alignment construction is performed based on word alignment tools. We refer to this method as weakly supervised because there are some supervised signals, but the accuracy is not 100%.

In this stage, although SVD is fast for training, its memory consumption increases as the amount of data increases. Therefore, we use gradient descent (GD) to optimize the mapping matrix W , seeking an approximate solution. The loss function is shown in Eq. (22), where N is the batch size.

$$\text{loss} = \frac{1}{|N|} \sum_{i=1}^N (Wx_i - y_i)^2 \quad (22)$$

3.2. Cross-lingual knowledge distillation

3.2.1. Knowledge transfer and semantic alignment

As shown in Fig. 1, in the second stage, we use the domain-specific pretrained model (PLM) of the source language as the teacher network T and a multilingual PLM as the student network S to perform cross-lingual semantic alignment and knowledge transfer. Specifically, given a parallel sentence pair between the source language (e.g., Chinese) and the target language (e.g., Mongolian) $\langle s_i, t_i \rangle$, we first input the Chinese sentence s_i into the teacher model T to obtain the word embeddings $h_T^{s_i}$. Using the mapping matrix W obtained from the first stage, we map the word embeddings to the target language space:

$$E_T^{s_i'} = E_T^{s_i} \times W \quad (23)$$

To obtain the sentence-level representation, we further apply the CLS pooling operation to the word embeddings, resulting in sentence representation as follows:

$$E_T^{\text{CLS}'} = \text{CLS}(E_T^{s_i'}) \quad (24)$$

Moreover, the parallel word pairs $\langle s_i, t_i \rangle$ are input into the S model to generate word embeddings $E_S^{s_i}$ and $E_S^{t_i}$, and the CLS pooling operation is applied to obtain:

$$E_S^{\text{CLS}} = \text{CLS}(E_S^{s_i}), \quad E_S^{\text{CLS}} = \text{CLS}(E_S^{t_i}) \quad (25)$$

We achieve knowledge transfer and alignment by minimizing the mean squared error (MSE) between the outputs of the teacher and student models, and the loss function is defined as:

$$\text{loss} = \frac{1}{|N|} \sum_{i=1}^N \left[(E_T^{\text{CLS}'} - E_S^{\text{CLS}})^2 + (E_T^{\text{CLS}'} - E_S^{\text{CLS}})^2 \right] \quad (26)$$

Mapping-then-pooling strategy: After word embeddings are processed through a mapping matrix, the semantic information of the target language can be more accurately reflected. This information is then aggregated via the [CLS] token of the PLM to improve the quality of sentence embeddings.

3.2.2. Model compression

Inspired by ALBERT (Lan, 2019), we design the smaller multilingual student model CLWKD_S by using parameter cycling and embedding bottleneck strategies. Parameter cycling: We initialize the smaller student model CLWKD_S with the parameters of the first M layers from CLWKD $RU = \{L_i | i \in [1, M]\}$, where L_i represents the i -th transformer layer. Embedding bottleneck: CLWKD, which is based on multilingual PLM training, usually requires large vocabularies V to support multiple languages, which leads to a very large number of parameters in the embedding layer. We add a bottleneck layer of size B between the embedding layer and the hidden layer H (Lan, 2019); thus, the size of the embedding layer is reduced from $O(V \times H)$ to $O(V \times B + B \times H)$. As shown in Fig. 5, we align the CLWKD and CLWKD_S embedding spaces via the loss function in Eq. (27), where $h_M^{s_i}, h_M^{t_i}$ denote the output of the CLWKD embedding layer and $h_S^{s_i}, h_S^{t_i}$ denote the output of the CLWKD_S bottleneck layer. After obtaining the smaller student model, we use a second-stage approach to transfer the domain knowledge from the teacher model to the smaller student model CLWKD_S.

$$\text{loss} = \frac{1}{|N|} \sum_{i=1}^N \left[(h_M^{s_i} - h_S^{s_i})^2 + (h_M^{t_i} - h_S^{t_i})^2 \right] \quad (27)$$

4. Experimentation

4.1. Data and settings

In the experiments, we selected Mongolian and Korean, both of which are agglutinative languages, as target languages, with high-resource Chinese as the source language. Mongolian, as a typical low-resource agglutinative language, is underrepresented in many

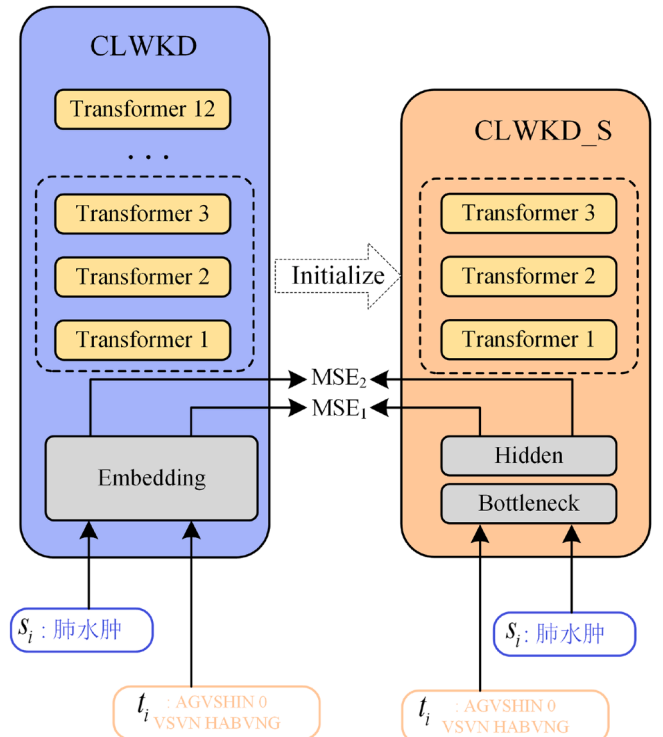


Fig. 5. Model compression.

Table 1
Distributions of the datasets.

Dataset	Units	Training set	Dev set	Test set	Total
CWMT	Sentence Pairs	66,000	1000	200	67,200
KCSP	Sentence Pairs	10,000	1000	200	11,200
MMCPT + CWMT	Sentence Pairs	1695 + 10000	213 + 200	213 + 200	2121 + 10400
CMLQA	Sentence Pairs	8000	1000	200	9200
CM-Edu	Sentence Pairs	1658 + 10000	208 + 200	207 + 200	2073 + 10400
CK-Med + KCSP	Sentence Pairs	4800 + 3200	600 + 200	600 + 200	6000 + 3600
MDNER	Sentences	3065	383	383	3831
MLNER	Sentences	1920	240	240	2400

mainstream multilingual pre-trained models (such as mBERT and XLM), making its domain semantic alignment tasks particularly challenging. To further evaluate the model's generalizability, we extended the experiments to Korean. Although Korean is considered a high-resource language in terms of pre-training corpora, it still lacks large-scale Korean-Chinese parallel corpora in specific domains, making this task equally representative and challenging.

In the first stage of CLWKD, we used general-domain parallel corpora for Mongolian-Chinese and Korean-Chinese, including publicly available datasets such as the CWMT and KCSP.³ In the second stage, we used parallel corpora from three domains: medical, legal, and educational. Since there is no publicly available Mongolian-Chinese medical domain dataset, we constructed our own Mongolian-Chinese medical terminology parallel dataset (MMCPT). For the legal domain, we used the publicly available CMLQA⁴ dataset. The educational domain data were sourced from Mongolian-Chinese abstracts of master's and doctoral theses from Inner Mongolia Normal University, and we built the Mongolian-Chinese educational parallel dataset (MC-Edu). For the Korean-Chinese medical domain, we used the publicly available Korean medical monolingual dataset KBMC⁵ and generated Chinese translations via machine translation to create the Korean-Chinese medical parallel dataset (CK-Med).

The performance of CLWKD was evaluated through three downstream tasks: cross-lingual word alignment, cross-lingual sentence retrieval, and cross-lingual entity recognition. The evaluation data for the word alignment and sentence retrieval tasks were selected from parts of the aforementioned corpora. For the entity recognition evaluation, we used the CCKS 2020 entity recognition dataset, along with the v1 version of the Mongolian medicine entity recognition dataset (MDNER) and a small-scale legal entity recognition dataset (MLNER), both constructed in our lab. Detailed information about all the datasets is provided in Table 1.

In the experiments, we used MC-BERT (Wang, 2023) as the multilingual PLM for low-resource languages. MC-BERT is a model fine-tuned on 150,000 Mongolian monolingual sentences and 67,200 Mongolian-Chinese parallel sentence pairs based on multilingual BERT.

In the adversarial training framework, the discriminator is a three-layer perceptron (MLP) with a hidden layer size of 2048. The dropout rate for the input layer is 0.1, whereas the dropout rate for the hidden layers is set to 0. The smoothing factor is 0.2, and the generator is updated every 5 steps. The batch size for both the discriminator and generator is 32, with a learning rate of 0.1 and a learning rate decay rate of 0.95. The adversarial training runs for 50 epochs. The alignment vector threshold is set to 0.002. In the refinement phase, the mapping matrix optimizer uses SGD with a learning rate of 0.1 trained for 20 epochs.

In the cross-lingual knowledge distillation phase, training is conducted for 100 epochs with a batch size of 32. A warmup rate of 0.1

is applied to stabilize early updates, using a learning rate of 16×10^{-5} . In the compression phase, the student model retains only 3 transformer layers.

4.2. Compared models

To evaluate the model's cross-lingual semantic alignment performance at different granularities, three cross-lingual evaluation tasks were designed: word alignment, bitext retrieval, and cross-lingual entity recognition. The comparison models included classic cross-lingual semantic alignment models, multilingual PLMs, recent fine-tuned multilingual PLMs, and knowledge distillation models.

Adv-SVD-CSLS (Conneau et al., 2017) and **Adv-SVD-NN** (Conneau et al., 2017): Based on GANs, these models are further trained by constructing aligned words and using the CSLS and NN methods to select nearest-neighbor word embeddings during evaluation tasks.

CLBT_SVD (Wang et al., 2019) and **CLBT_GD** (Wang et al., 2019): These methods first obtain contextually aligned word pairs via fast_align and then compute contextual word embeddings via BERT. The mapping matrix W is then learned via SVD and GD.

mBERT (Kenton & Toutanova, 2019): A multilingual BERT model released by Google, supporting over 100 languages and demonstrating strong cross-lingual transfer capabilities.

CINO (Yang et al., 2022): This method continues training XLM-R on corpora of various minority languages, enhancing its understanding of Tibetan, Mongolian, Uyghur, Kazakh, Korean, Zhuang, Cantonese, and other minority languages and dialects.

InfoXLM (Chi et al., 2020): This method continues training XLM-R by masked language modeling, translation language modeling and translation ranking tasks.

XLM-Align (Chi et al., 2021): This method performs self-supervised word alignment on parallel sentences and trains by randomly masking some tokens in bilingual text pairs.

DAP (Li et al., 2023): This is a state-of-the-art cross-lingual sentence embedding model that uses token-level information.

AoE (Li & Li, 2024): Enhances the precision of semantic representations by optimizing angular differences in complex spaces and mining text similarities within saturated regions.

mKD (Reimers & Gurevych, 2020): This model learns multilingual sentence embeddings via cross-lingual knowledge distillation.

msKD (Ding et al., 2022): This method trains small cross-lingual models via multistage knowledge distillation.

IMASCL (Park et al., 2024): It primarily guides the student model to learn relationships between cross-lingual sentences by aligning the monolingual sentence similarity structures of the teacher model.

All the comparison models, except for CINO, were trained on our datasets.

4.3. Word alignment

On the basis of previous work (Chi et al., 2021), word-level semantic alignment performance was evaluated via word alignment. Word alignment involves identifying corresponding word pairs within parallel sentences.

³ KCSP is a Korean-Chinese parallel corpus obtained from the OPUS project (<https://opus.nlpl.eu>)

⁴ <http://cdata.org/en/p/1061/>

⁵ <https://huggingface.co/datasets/SungJoo/KBMC>

Table 2
Evaluation results for word alignment.

Model	Weakly Supervised			Unsupervised		
	P	R	F1	P	R	F1
Adv_SVD_CSLS	54.97	59.05	56.94	42.91	45.71	44.27
_NN	53.41	57.32	55.30	36.94	39.39	38.13
CLBT_SVD_CSLS	55.37	59.45	57.34	–	–	–
_GD_CSLS	60.80	65.47	63.05	–	–	–
CINO	51.99	51.43	51.71	–	–	–
InfoXLM	51.54	45.32	48.23	–	–	–
XLM-Align	45.82	39.43	42.39	–	–	–
CLWKD_SVD_CSLS	68.25	57.56	62.45	–	–	–
_GD_CSLS	71.96	60.00	65.43	47.80	43.01	45.28
_GD_NN	69.30	54.76	61.18	41.85	36.96	39.25

Table 2 shows the precision, recall and F1 scores for word alignment in both languages. In weakly supervised learning, our CLWKD_GD_CSLS model outperformed the classic Adv_SVD_CSLS and Adv_SVD_NN models, with F1 improvements of 8.49 % and 10.13 %, respectively, in Mongolian-Chinese word vector alignment, demonstrating the superiority of our proposed method over traditional models. Compared with the CLBT_GD_CSLS model, our CLWKD_GD_CSLS model showed a 2.38 % improvement in the evaluation task and a 5.11 % improvement over the CLBT_SVD_CSLS model, indicating that the generative adversarial network (GAN) effectively reduces the distance between the Mongolian and Chinese vector spaces. Compared with the pre-trained CINO model, the SVD method used in CLWKD showed a 10.74 % improvement, whereas the GD method resulted in a 13.72 % improvement. Furthermore, compared with the fine-tuned InfoXLM and XLM-Align models, the CLWKD_GD_CSLS method yielded enhancements of 17.2 % and 23.04 %, respectively. These comparison results fully demonstrate that weakly supervised and self-supervised training methods that rely solely on limited data cannot achieve the desired results when the multilingual XLM-R large model is used. In contrast, CLWKD significantly improves performance in cross-lingual word alignment tasks through more effective alignment methods and optimization strategies. This suggests that in multilingual tasks, appropriate alignment techniques and advanced optimization methods are crucial for fully realizing model potential, especially when data are limited.

Compared with the baseline model, our unsupervised learning models yielded improvements of 1.01 % and 6.03 %. In addition, among all the models, the CSLS method consistently outperformed the NN method in finding aligned word vectors, with the F1 score being higher by up to 6.03 %.

In the experiments, we also attempted to introduce the CycleGAN cyclic consistency constraint strategy. However, during training, we frequently encounter gradient vanishing or gradient explosion issues, making it difficult for the model to converge stably. CycleGAN relies on the cycle consistency loss of “ $A \rightarrow B \rightarrow A$ ” and “ $B \rightarrow A \rightarrow B$ ” to ensure reversible transformations. However, owing to significant differences in syntax, word order, and morphology between Mongolian and Chinese, maintaining a one-to-one correspondence when transforming back is challenging, thus causing the cycle consistency constraint to fail. In contrast, the standard GAN optimizes only the distribution alignment of cross-lingual mappings, making it potentially more stable for text alignment tasks. CycleGAN was originally designed for style transfer tasks, but owing to the large structural differences between Mongolian and Chinese, its effectiveness in unsupervised alignment may be limited. Additionally, CycleGAN is sensitive to data distribution consistency, making it prone to optimization difficulties or mode collapse in low-resource scenarios. Although we attempted to combine parallel data and word pairs for training, achieving stable mapping results was difficult.

Table 3 presents the word alignment results for the Mongolian-Chinese legal and educational domain datasets. It is evident that the

performance of mBERT + W is significantly lower; on the one hand, this is due to the limited coverage of Mongolian during the mBERT pretraining process (we used the fine-tuned version, while the original mBERT model hardly covered Mongolian). On the other hand, the mapping matrix W is the result of CLWKD’s first-stage training, which was trained solely on the general Mongolian-Chinese dataset and does not include domain-specific data, thus limiting its domain-specific word alignment capability.

After introducing a small amount of domain data, further training of CLWKD significantly outperforms the fine-tuned InfoXLM and XLM-Align models. The Avg_F1 score improved by 9.26 % and 8.41 % in the legal and educational domains, respectively. The DAP model, which also performs token-level alignment during fine-tuning, shows competitive performance in word alignment tasks: the Avg_F1 score improved by 0.23 % in the legal and educational domains. This finding indicates that token alignment methods have a positive effect on improving word alignment performance. Compared with the state-of-the-art text embedding similarity model AoE, CLWKD achieved an Avg_F1 improvement of 0.14 % in the legal and educational domains; this suggests that CLWKD is more suitable for fine-grained semantic alignment tasks such as word alignment, where weak supervision via parallel sentences is employed. Moreover, the experimental results vary when different domain models are used as teacher networks in CLWKD, indicating that the performance of the teacher model itself significantly affects the alignment results. A detailed analysis of this will be provided in Section 4.4, in conjunction with sentence retrieval performance.

To evaluate the model’s cross-lingual word alignment capability on other low-resource agglutinative languages, we conducted word-level alignment experiments on the smaller Korean-Chinese dataset. The experimental results are shown in Table 4. Since mBERT’s pretraining covers both Korean and Chinese, its performance on this task is significantly better than the Mongolian-Chinese word alignment results presented in Table 3. However, the mBERT pretraining process involves training on a large amount of multilingual data without explicitly utilizing semantic alignment information, which limits its alignment performance. When we trained the cross-lingual semantic alignment mapping matrix on general-domain parallel corpora and applied it to mBERT, the model’s word alignment performance improved significantly in both the general and medical domains. Building upon this, the CLWKD model, obtained through knowledge distillation using cross-lingual domain data, demonstrated even greater advantages in both domains. Compared with the InfoXLM, XLM-Align, DAP, and AoE models fine-tuned with domain data, CLWKD achieved the highest F1 score improvements of 12.91 %, 9.22 %, 2.4 %, and 3.55 % in the medical domain, respectively, and 4.26 %, 2.13 %, 4.06 %, and 3.55 % in the general domain, respectively. These experimental results indicate that combining cross-lingual semantic alignment mapping with domain knowledge distillation can effectively address the shortcomings of pretrained models in word-level semantic alignment, significantly improving word alignment accuracy.

Overall, domain-specific semantic alignment methods primarily use sentence-level alignment as the main training objective, resulting in relatively weaker performance on word-level alignment. To more comprehensively evaluate the model’s alignment capability, we next focus on analyzing its performance in the domain sentence alignment task and further analyze the performance of the two stages of CLWKD.

4.4. Bitext retrieval

Based on previous works (Artetxe & Schwenk, 2019; Chi et al., 2020; Feng et al., 2020), we evaluate the sentence-level semantic alignment capability via the dual-text retrieval task. The evaluation metrics include the average retrieval accuracy from the source language to the target language (s2t) and from the target language to the source language (t2s). Tables 5 and 6 present the sentence retrieval performance of CLWKD on

Table 3

Evaluation results of word alignment on the Mongolian-Chinese datasets in the legal and education domains.

Model	Teacher Model	Law			Education			Avg_F1
		P	R	F1	P	R	F1	
mBERT + W		15.30	16.52	15.89	14.02	17.13	15.42	15.66
InfoXLM		34.47	36.25	35.35	47.96	51.87	49.84	42.60
XLM-Align		36.65	46.19	40.87	40.09	44.02	46.02	43.45
DAP		50.44	50.90	50.67	50.79	54.50	52.58	51.63
AoE		45.85	54.87	49.94	51.48	55.66	53.48	51.72
CLWKD	LegalRoBERTa/TAL-EduBERT	46.15	54.93	50.16	51.41	55.88	53.55	51.86
CLWKD	RoBERTa-wwm-ext	44.44	52.85	48.28	49.14	53.49	51.22	49.75
CLWKD	Zh-BERT	45.44	54.22	49.44	49.13	53.11	51.04	50.24
CLWKD	CKIP BERT	45.50	53.98	49.38	–	–	–	–
CLWKD	Lawformer	45.66	54.26	49.59	–	–	–	–

Table 4

Evaluation results of word alignment on the Korean-Chinese cross-lingual datasets.

Model	Teacher Model	Medical			General			Avg_F1
		P	R	F1	P	R	F1	
mBERT		30.31	30.11	30.21	28.07	27.86	27.96	29.09
mBERT + W		23.44	47.33	31.35	37.22	42.57	39.72	35.54
InfoXLM		40.30	39.90	39.75	35.72	36.33	36.02	37.89
XLM-Align		42.84	42.60	42.72	38.75	37.56	38.15	40.44
DAP		49.65	49.43	49.54	36.00	36.44	36.22	42.88
AoE		38.42	38.08	38.25	43.92	43.74	43.83	41.04
CLWKD	MedBERT	50.40	50.27	50.33	33.66	31.36	32.47	41.40
CLWKD	ERNIE-Health	53.09	50.83	51.94	39.70	40.88	40.28	46.11
CLWKD	BioBERT	47.06	46.87	46.96	32.22	29.20	30.64	38.80
CLWKD	MC-BERT	44.91	44.72	44.81	32.76	29.99	31.31	38.06

Table 5

Evaluation results of bilingual bitext retrieval on Mongolian-Chinese and Korean-Chinese datasets across medical and general domains.

Model	Teacher	Mongolian-Chinese				Korean-Chinese					
		Medical		General		Medical		General		Avg_s2t	Avg_t2s
		s2t	t2s	s2t	t2s	s2t	t2s	s2t	t2s		
mBERT		–	–	–	–	12.77	10.83	14.00	12.50	13.39	11.67
mBERT + W		10.63	8.92	16.83	10.89	15.56	14.30	16.83	10.89	14.96	11.25
XLM-Align		15.09	14.24	27.14	24.12	17.00	15.00	20.21	17.02	19.86	17.06
InfoXLM		16.04	15.47	26.13	25.53	19.00	16.50	15.96	20.74	19.28	19.56
DAP_cls		70.61	69.19	65.45	65.69	75.84	74.15	68.59	67.29	70.12	69.08
DAP_avg		69.66	68.72	64.47	64.96	73.44	71.83	67.65	64.82	68.81	67.58
AoE		33.49	36.42	70.67	74.40	38.30	40.83	53.72	54.50	49.05	51.54
IMASCL	mE5	24.83	27.25	31.90	35.74	83.74	88.09	73.37	77.33	53.46	57.10
mKD	MedBERT	77.47	78.26	66.43	67.42	85.83	83.33	48.60	47.10	69.58	69.03
MsKD	MedBERT	73.14	74.23	65.44	65.93	82.00	80.5	46.3	45.10	66.72	66.44
CLWKD	MedBERT	81.13	79.24	69.66	70.63	90.33	89.83	51.34	49.85	73.12	72.39
CLWKD	ERNIE-Health	76.88	75.94	66.74	63.10	76.83	72.00	35.24	33.97	63.92	61.25
CLWKD	BioBERT	14.56	12.13	–	–	60.50	57.00	37.43	34.92	37.50	34.68
CLWKD	MC-BERT	72.16	67.45	53.88	50.97	84.66	84.16	42.91	39.92	63.40	60.63
CLWKD_S	MedBERT	74.52	75.47	66.26	65.04	84.5	82.5	47.90	46.80	68.30	67.45

Mongolian-Chinese and Korean-Chinese language pairs in the medical, general, legal, and educational domains.

Compared with classic multilingual PLM models, [Tables 5 and 6](#) show that CLWKD significantly improves both Avg_s2t and Avg_t2s in the medical and general domains for both Mongolian-Chinese and Korean-Chinese language pairs, as well as in the legal and educational domains for Mongolian-Chinese, compared with models such as mBERT, mBERT + W, XLM-Align, and InfoXLM. This finding indicates that multilingual models trained on large-scale general-domain corpora, when fine-tuned with only a small amount of domain-specific data in low-resource settings, struggle to achieve good domain-specific sentence-level semantic alignment. Although mBERT + W shows improvements in word-level alignment, its mapping matrix W is constructed based on general-domain corpora, with the optimization objective focused on

word alignment, which limits its performance on sentence-level semantic alignment tasks for specific domains.

Compared with the recent sentence embedding semantic alignment models DAP and AoE, [Table 5](#) shows that CLWKD improves the retrieval accuracy of Avg_s2t and Avg_t2s by 2.99 % and 3.31 %, respectively, when compared with DAP and by 24.07 % and 20.85 %, respectively, when compared with AoE, in the medical and general domains for both language pairs. Notably, CLWKD shows a particularly significant improvement in the medical domain compared with AoE, whereas the advantage in the general domain is relatively smaller, indicating its greater advantage under limited domain data conditions. Furthermore, as shown in [Table 6](#), in the legal domain, owing to the relatively large scale of the Mongolian-Chinese legal dataset, AoE's performance is close to that of CLWKD, demonstrating some competitiveness but also

Table 6

Evaluation results of bilingual bitext retrieval on Mongolian-Chinese datasets across the law and education domains.

Model	Teacher	Law		Education		Avg_s2t	Avg_t2s
		s2t	t2s	s2t	t2s		
mBERT + W		6.00	9.50	12.43	10.89	9.21	10.19
XLM-Align		20.00	21.30	37.27	33.64	28.63	27.47
InfoXLM		16.50	17.45	34.55	31.82	25.525	24.63
DAP_cls		74.20	77.13	50.96	51.44	62.58	64.28
DAP_avg		73.21	76.90	50.17	50.25	61.69	63.57
AoE		86.05	84.20	50.76	53.75	68.40	68.97
IMASCL	mE5	26.47	25.36	24.82	22.54	25.64	23.95
mKD	Lawformer/TAL-EduBERT	80.10	79.20	48.89	44.71	64.49	61.95
MsKD	Lawformer/TAL-EduBERT	76.80	73.70	47.66	43.98	62.23	58.84
CLWKD	Lawformer/TAL-EduBERT	87.00	87.70	50.63	51.47	68.81	69.58
CLWKD	CKIP BERT	32.00	33.00	–	–	–	–
CLWKD	LegalRoBERTa	85.90	84.30	–	–	–	–
CLWKD	RoBERTa	81.7	80.2	50.12	48.64	65.91	64.42
CLWKD	Zh-BERT	70.02	69.13	35.38	31.94	52.7	50.53

reflecting its sensitivity to the data scale. Finally, as shown in Table 6, CLWKD outperforms DAP and AoE by 6.23 % and 5.3 %, respectively, in Avg_s2t, and by 0.41 % and 0.61 %, respectively, in Avg_t2s, in the legal and educational domains. These results demonstrate that the dual-stage training strategy employed by CLWKD provides significant advantages in low-resource domain tasks.

Compared with cross-lingual knowledge distillation models, particularly the recently proposed IMASCL method, Tables 5 and 6 show that CLWKD achieves significantly higher performance across three Mongolian-Chinese domain datasets. IMASCL primarily guides the student model to learn cross-lingual semantic relations by aligning sentence similarity structures from the teacher model; however, due to the substantial syntactic and morphological differences between Mongolian and Chinese, relying solely on similarity alignment fails to capture the complex semantic mappings between the two languages. Moreover, we observed that IMASCL heavily relies on sentence embedding-based PLMs. When replacing the teacher model from mE5⁶ to mBERT, the performance drop became more pronounced, as shown in the comparison experiments under the 5.8 high-resource setting. This reliance makes IMASCL less suitable for low-resource domain scenarios, where it is typically necessary to leverage both sentence-level and word-level alignment data to compensate for data scarcity. Compared to mKD, Tables 5 and 6 show that CLWKD, with MedBERT as the teacher model, improves Avg_s2t and Avg_t2s by 3.53 % and 3.36 %, respectively, in the medical and general domains for both language pairs. In the Mongolian-Chinese legal and educational domains, with Lawformer and TAL-EduBERT as teacher models, Avg_s2t and Avg_t2s improve by 4.32 % and 7.63 %, respectively. These results indicate that the cross-lingual mapping matrix W , trained on a general-domain dataset, not only enhances the model's performance in specific domains but also significantly improves its applicability and transferability in broader tasks.

Compared with the recent cross-lingual knowledge distillation compression model MsKD, Tables 5 and 6 show that the compressed CLWKD_S, with MedBERT as the teacher model, improves Avg_s2t and Avg_t2s by 1.58 % and 1.01 %, respectively, in the medical and general domains for both language pairs. These results validate the effectiveness and robustness of the “cross-lingual mapping matrix + cross-lingual knowledge distillation” dual-stage strategy in small model scenarios, demonstrating that this method retains strong cross-lingual semantic transfer capabilities even in parameter-limited compressed settings.

Impact of Different Teacher Models on CLWKD. In the medical domain, CLWKD performs best in the word alignment task when ERNIE-

Health is used as the teacher model, whereas MedBERT achieves better results in the sentence retrieval task. During pre-training, ERNIE-Health introduces an entity-level masking strategy that masks key medical entities such as disease names, medications, and symptoms. This strategy guides the model to focus on the word-level representation and semantic boundary recognition of medical terms, thereby enabling more targeted transfer of entity-level semantic information to the student model during knowledge distillation. As a result, CLWKD performs best in the word alignment task when using ERNIE-Health as the teacher model. In contrast, MedBERT is pre-trained on large-scale clinical text, including numerous clinical sentences, physician notes, and patient complaints. Its training objectives include random token masking and sentence order prediction, which help enhance the model's ability to capture sentence-level semantic structures. Therefore, MedBERT is better at modeling contextual logic and domain-specific concepts at the sentence level, and achieves better performance in the sentence retrieval task within the CLWKD framework. Both models are pre-trained on large-scale Chinese medical corpora, giving them a significant advantage in semantic transfer for the medical domain. In contrast, although BioBERT is also a model for the medical domain, its training data mainly consist of English text, resulting in weaker performance on Mongolian-Chinese and Korean-Chinese parallel corpora. However, in the model generalization experiments in Section 5.8, it performs well on the high-resource English-Chinese dataset, indicating that its transfer capability is better suited to languages that are closer or have sufficient resources.

In the legal domain, Tables 3 and 6 show that LegalRoBERTa and Lawformer, with deep pretraining on Chinese legal texts, significantly outperform other models. The rich legal knowledge embedded in these models is effectively transferred to low-resource languages through cross-lingual distillation. In the educational domain, TAL-EduBERT also helps CLWKD achieve optimal performance due to its task alignment and specialized training. In contrast, CKIP-BERT, which is trained only on Chinese legal texts, performs poorly in distillation due to the limited data scale and diversity. General models such as RoBERTa and Zh-BERT also show a significant decrease in performance on specialized domain datasets. These results suggest that the match between the teacher model's language coverage and domain adaptability directly determines the effectiveness of knowledge transfer and is a key factor influencing cross-lingual distillation performance.

Impact of Domain Data Characteristics on Model Transfer Alignment Capability. We identified that, in addition to the model architecture and teacher model, the language structure and expression style of different domain data significantly affect cross-lingual sentence alignment performance. Taking the legal domain as an example, this type of data primarily consists of question-and-answer-style legal documents,

⁶ <https://huggingface.co/intfloat/multilingual-e5-large>

Table 7
Evaluation results for NER on the medical and legal datasets.

Model	Medical			Law			Avg_F1
	P	R	F1	P	R	F1	
mBERT	60.90	60.84	60.87	50.29	50.82	50.55	55.71
mBERT-BiGRU	61.86	59.20	60.50	52.26	51.33	51.79	56.15
mBERT-BiLSTM	59.34	58.91	59.12	49.15	50.86	49.99	54.56
DAP_cls	71.11	69.39	70.24	55.48	56.44	55.96	63.10
AoE	70.18	68.35	69.52	58.41	57.61	58.01	63.63
mKD	70.46	68.86	69.65	55.30	59.05	57.11	63.38
MsKD	66.9	63.14	64.97	50.82	52.76	51.77	58.37
CLWKD (Stage 1)	70.79	69.71	70.25	60.36	58.94	59.64	64.95

which have clear sentence structures and a high proportion of general vocabulary. The language structure is closer to that of general-domain corpora, making it more conducive to the transfer application of the general-domain mapping matrix, thus enabling CLWKD to perform relatively stably in this domain. The data in the educational domain are obtained from abstracts of master's and doctoral theses in education at Inner Mongolia Normal University. The overall structure is highly consistent, and the themes are concentrated. Although a random sentence selection strategy was used during training to increase sample diversity, the strong semantic similarity and thematic consistency between the texts limited the model's ability to recognize semantic differences between sentences. In the medical domain, the dataset used is the Mongolian-Chinese medical domain parallel terminology dataset, which is highly standardized and specialized in terms of terminology; this helps the model align specific semantic units, leading to superior performance in this domain.

These results indicate that differences in sentence structure and semantic span of the corpus have a significant effect on sentence-level semantic modeling capability. CLWKD performs better on domain corpora with higher degrees of structural and terminological consistency.

4.5. Cross-lingual named entity recognition

To evaluate the model's performance on cross-lingual named entity recognition tasks, we conducted an experimental analysis using Mongolian-Chinese medical and legal domain datasets. First, we fine-tuned the cross-lingual alignment model via the CCKS 2020 Chinese medical named entity recognition dataset. We then incorporated cross-lingual prompting techniques, where Chinese prompt information guided the entity recognition process for Mongolian medical and legal domain texts, followed by further training and testing. We believe that if the model has already developed strong Mongolian-Chinese semantic alignment during pretraining, the Chinese prompts can provide clear semantic cues for low-resource languages, thereby significantly improving recognition performance.

As shown in Table 7, CLWKD (Stage 1) achieved average F1 improvements of 9.23 %, 8.80 %, and 10.39 % in entity recognition tasks across the two domains compared with multilingual baseline models (mBERT, mBERT-BiGRU, and mBERT-BiLSTM). Compared with the AoE model, CLWKD achieved a 1.32 % average F1 improvement; compared with the traditional knowledge distillation model mKD, the improvement was 1.56 %. Although DAP performed worse than did CLWKD sentence retrieval tasks, it showed strong competitiveness in entity recognition tasks. This is especially evident on medical datasets. Its strategy of enhancing language unit mapping through subword alignment helps improve the model's sensitivity to fine-grained entities. We analyze this phenomenon further in Section 5.6.

Overall, the multigranularity mapping matrix generated in Stage 1 of CLWKD (pretrained model + mapping structure) demonstrates stronger representational capacity in cross-lingual word alignment and entity recognition tasks, reflecting its significant advantage in fine-grained semantic alignment. In contrast, Stage 2 exhibits greater stability and

cross-lingual semantic capturing ability, making it more suitable for macrolevel semantic modeling in sentence-level alignment tasks.

5. Analysis

5.1. Different word segmentation methods for cross-lingual semantic mapping

To alleviate the data sparsity problem, this study incorporates the morphological characteristics of agglutinative languages and adopts a morphological decomposition approach. Table 8 presents experimental results on Mongolian as a representative agglutinative language. In the table, "Root" denotes retaining only the root form, "Seg" refers to segmenting the affixes, and "Cut" indicates directly removing the affixes.

Table 8 shows that the best results are obtained by removing affixes, regardless of whether the method is weakly supervised or unsupervised. In the weakly supervised experiments, removing affixes yielded an average F1 score that was 3.8 % higher than that when segmenting affixes and 17.67 % higher than that when retaining only the root. The analysis revealed that when only the root is retained, the word alignment accuracy when fast_align is used is relatively low, and the average similarity between words during training is also low, resulting in a lower alignment rate in the vector space. Although the root retains the most basic information of the word, it is difficult to align its meaning with Chinese in context, and the semantic information is still insufficient.

In unsupervised learning, when only the root is retained, the model often encounters issues such as gradient vanishing or gradient explosion. Despite the root retaining the basic information of the word, the lack of contextual information makes it difficult to express the semantics when aligning with Chinese. In practice, even though attempts have been made to mitigate these issues by adding thresholds when training the model with the root, the results are still unsatisfactory. These findings further validate the challenges of fine-tuning GAN networks with small samples under unsupervised conditions.

5.2. Ablation experiments for cross-lingual semantic mapping

The purpose of this experiment was to conduct an ablation analysis to evaluate the contributions of different modules in the first stage of the model. Specifically, we set the weights of the model's pre-trained generator (Pre_Mapping), optimized mapping (Opt_mapping), training threshold (train_threshold), and testing threshold (test_threshold) to zero and then compared the performance with that of the original model. Table 9 presents the results of the ablation study on cross-lingual mapping conducted on the Mongolian-Chinese general-domain dataset.

As shown in the 4th row of Table 9, the Opt_mapping module plays a crucial role in the model; removing Opt_mapping resulted in a direct decrease in the F1 score of 33.13 %. The 5th row of Table 9 indicates that the pre-trained generator also has a significant effect on the overall model. When this module was removed, the F1 score decreased by 3.83 % on the evaluation task. Pre-training the generator not only stabilizes the training process of the GAN but also enhances the overall model performance. The 6th row of the table shows that removing the training threshold led to a decrease of 2.55 % in the F1 score, whereas

Table 8
Different word slicing methods.

Model	Weakly Supervised			Unsupervised		
	P	R	F1	P	R	F1
CLWKD_SVD_Cut	68.25	57.56	62.45	—	—	—
CLWKD_GD_Cut	71.96	60.00	65.43	47.80	43.01	45.28
CLWKD_SVD_Root	36.49	40.80	38.52	—	—	—
CLWKD_GD_Root	53.70	54.34	54.02	—	—	—
CLWKD_SVD_Seg	60.26	57.58	58.89	—	—	—
CLWKD_GD_Seg	63.46	59.42	61.38	35.15	32.48	33.76

Table 9
Ablation analysis of cross-lingual mapping.

Model	P	R	F1
Baseline	54.97	59.05	56.94
Full system	71.96	60.00	65.43
- Opt_mapping	34.18	31.88	32.30
- Pre_Mapping	59.62	63.74	61.61
- train_threshold	60.62	65.30	62.88
- test_threshold	70.88	58.19	63.91

removing the testing threshold resulted in a 1.52 % decrease. This suggests that setting thresholds is critical for model performance, as it helps filter out noise or incorrect correspondences, retaining only the correct correspondences.

5.3. Ablation analysis of cross-lingual knowledge distillation

The ablation study included experiments that removed the cross-language mapping matrix W from CLWKD, examined the effects of the order of mapping and pooling, compared average pooling and [CLS] pooling results, and analyzed the impact of removing the bottleneck layer from the compressed CLWKD_S. During model compression, the mapping matrix was not used, as CLWKD_S was initialized with the bottleneck layer and embedding layer from CLWKD. Table 10 presents the results of cross-lingual knowledge distillation experiments conducted on the Mongolian-Chinese medical and general-domain datasets.

When training the CLWKD model using only domain-specific data, the limited amount of such data led to overfitting on both the medical and general-domain test sets. This finding indicates that the data constraints hinder the model's ability to learn broadly applicable features, thereby reducing its performance across different tasks and domains.

Table 10 shows that removing the general domain cross-language mapping matrix from CLWKD resulted in decreases of 2.47 % and 0.17 % on the s2t and t2s tasks, respectively, on the medical dataset and decreases of 1.22 % and 1.7 %, respectively, on the general dataset. This result highlights the significant role of the cross-language mapping matrix W in enhancing the model's domain alignment performance.

Pooling after mapping (i.e., [CLS] pooling) led to a 1.93 % average decrease in performance across both datasets compared with mapping after pooling. This suggests that mapping first retains fine-grained relationships between word embeddings, thereby enhancing sentence alignment performance through preserved word alignment capabilities.

Additionally, using CLS pooling after mapping outperforms average pooling by an average of 0.77 % on the sentence retrieval tasks across the two datasets. These results demonstrate that [CLS] pooling is more effective in capturing sentence-level semantic information and improving performance in cross-language tasks.

Removing the bottleneck layer from CLWKD_S led to a significant drop in the model's cross-lingual sentence-level semantic alignment performance, with reductions of 8.02 % and 21.96 % on the medical and

Table 10
Ablation analysis of cross-lingual knowledge distillation. OVF stands for overfitting.

Model	Mapping or Pooling	MMCPT		CWMT	
		s2t	t2s	s2t	t2s
CLWKD (MMCPT)	Mapping + Average Pooling	OVF	OVF	OVF	OVF
CLWKD (MMCPT)	Mapping + CLS Pooling	OVF	OVF	OVF	OVF
CLWKD-W	Mapping + CLS Pooling	78.66	79.07	68.44	68.93
CLWKD	CLS Pooling	78.13	78.54	67.63	68.63
CLWKD	Mapping + Average Pooling	79.71	78.77	68.44	70.63
CLWKD_S	—	74.52	75.47	66.26	65.04
CLWKD_S -bottle	—	67.45	66.50	44.66	42.71
CLWKD(all)	Mapping + CLS Pooling	81.13	79.24	69.66	70.63

general datasets, respectively. This highlights the crucial role of the bottleneck layer in preserving the model's semantic integrity.

5.4. Impact of dataset size

The impact of training dataset size on the performance of unsupervised cross-lingual mapping. To improve model performance, we attempted to train the GAN with more data. When 360,000 Mongolian-Chinese sentence pairs were used, we observed increases in both unsupervised evaluation metrics, as shown in Fig. 6. This finding indicates that unsupervised training benefits from larger datasets. However, when the alignment matrix was optimized with 360,000 sentence pairs, the highest alignment accuracy achieved was 52 %. In contrast, using 67,000 sentence pairs yielded an alignment accuracy of 58 %. Neither approach matched the effectiveness of the alignment achieved with the 67,000 sentence pairs when adversarial training and matrix optimization were applied. Our analysis revealed that the quality of the 360,000 Mongolian-Chinese sentences was not high; many contained nonstandard characters, and the Latin transliteration was not manually preprocessed. This highlights the importance of data quality.

Impact of Training Data Size on Cross-lingual Knowledge Distillation Performance. As shown in Fig. 7, with increasing data size, the model's retrieval accuracy in both the s2t and t2s directions improves significantly. The convergence trend shows that models with larger data sizes (e.g., s2t_15, s2t_20) quickly improve in the first 20 epochs and converge earlier. However, models with smaller data sizes struggle to effectively learn semantic alignment features, even when trained for 100 epochs. We found that medium-sized datasets (e.g., s2t_8, s2t_10) already achieve near-optimal performance in both retrieval directions, and the training process is stable and reliable; this indicates that in practical applications, large-scale datasets are not necessary to achieve efficient and effective knowledge distillation.

5.5. Data augmentation

This section explores two data augmentation methods to alleviate the “lack of parallel corpora” problem. We introduced two data augmentation strategies, back-translation (BT) and machine translation (MT), on the training set of the Mongolian-Chinese medical dataset. Back-translation generates semantically equivalent but diversely expressed sentence pairs by translating Chinese sentences into Mongolian and then back to Chinese. Machine translation involves translating the CCKS Chi-

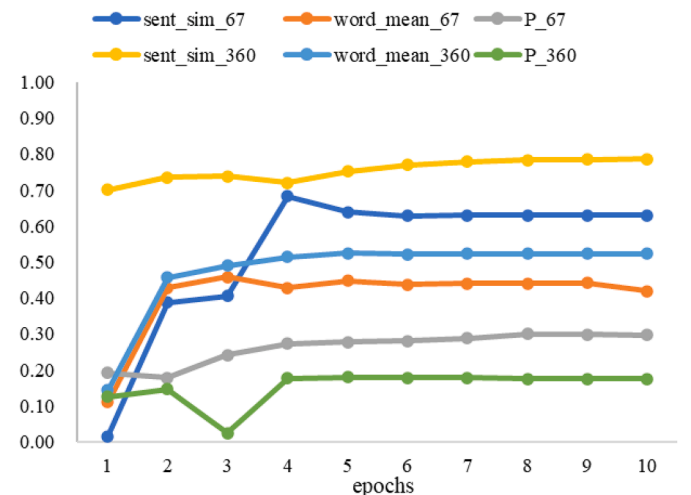


Fig. 6. Impact of training data size on the performance of unsupervised cross-lingual mapping, where the number of sentence pairs is measured in thousands.

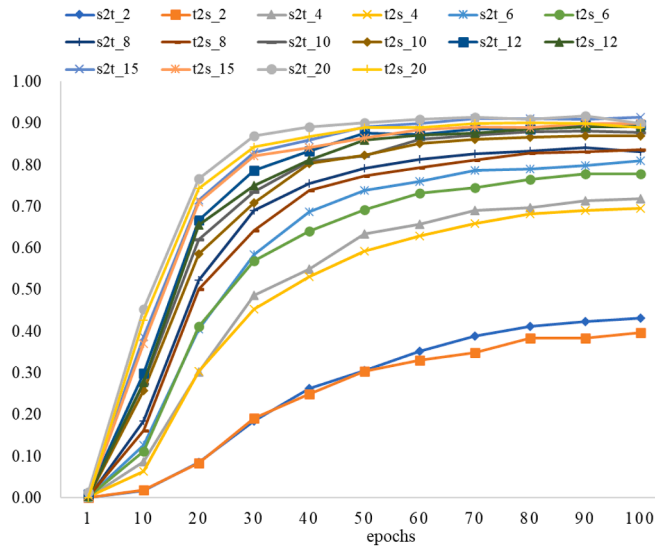


Fig. 7. Impact of training data size on the performance of cross-lingual knowledge distillation, where the number following the underscore (“_”) indicates the corresponding number of training sentence pairs (in thousands).

nese medical domain text into Mongolian, generating sentence pairs with actual semantic context.

The experimental results, shown in Fig. 8, indicate that both augmentation strategies outperform the original terminology data (OR), effectively improving the model’s training and alignment performance. Among them, back-translation (BT_s2t and BT_t2s) shows the best performance in terms of both convergence speed and final accuracy, with a significant improvement, especially between epochs 10–40. This demonstrates that by introducing structurally diverse but semantically consistent training samples, back-translation effectively enhances the model’s robustness to cross-lingual expression differences. Machine translation (MT_s2t and MT_t2s) also outperforms the original data group, indicating its ability to expand data diversity and improve the model’s generalizability; however, its performance improvement is slightly lower than that of back-translation, which may be attributed to the following reasons. First, the translation corpus comes from real medical sentences, whereas the original training data consists of terminology-based parallel sentences, which differ in sentence structure style; second, the complete medical sentence structure is more complex and semantically noisy, and machine translation quality fluctuates, making it prone to errors or amplifying noise, thus affecting the augmentation effect.

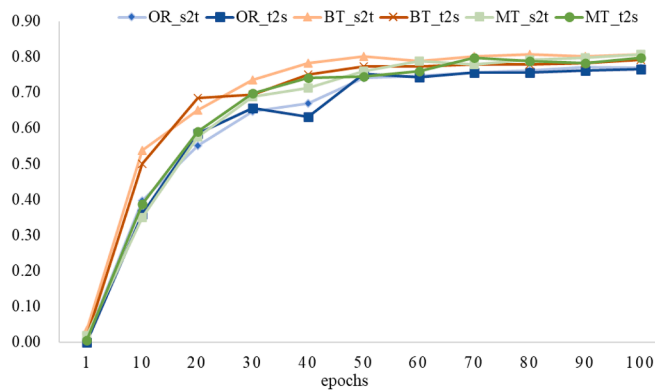


Fig. 8. Impact of back-translation and machine translation on Mongolian-Chinese sentence alignment performance. The evaluation metrics are sentence retrieval accuracies in both the s2t and t2s directions across different epochs.

Table 11

Case study analysis of cross-lingual NER.

Context	SINJI TEMDEG : D0TOR BACIMDAHV, VCAGARDAHV, HEBTEBEL BOSHVR DVR_A UGEI, CISVN DARVLTA NI D0OR_A, SOLIYARAJV GUYUHU JERGE SINJI-TEY.
Answer	D0TOR BACIMDAHV, VCAGARDAHV, CISVN DARVLTA NI D0OR_A, SOLIYARAJV GUYUHU
CLWKD	CISVN DARVLTA NI D0OR_A, D0TOR BACIMDAHV, SOLIYARAJV GUYUHU
DAP	CISVN DARVLTA NI D0OR_A, D0TOR BACIMDAHV, SOLIYARAJV GUYUHU
MsKD	CISVN DARVLTA NI D0OR_A, D0TOR BACIMDAHV

5.6. Case study

To explore the generalizability of CLWKD, we conducted a case study using neurology case data. The training set did not include neurology case data and consisted mainly of inpatient data from the respiratory department.

Table 11 shows that the two models trained with token alignment information, CLWKD and DAP, were able to identify the entity “SOLIYARAJV GUYUHU” (translated as “crazy running”), which was not present in the training set, whereas the other models failed to do so. This further demonstrates that token alignment models have a stronger ability to handle the fine-grained features and patterns of language. By breaking words into smaller token units, token alignment models can better capture subtle differences and details in language, enabling recognition of even unseen vocabulary through the combination of existing tokens. This fine-grained alignment mechanism is more effective at addressing complex morphological phenomena, such as affix changes and word form reduction, thus improving the accuracy of entity recognition.

To further validate this performance, experiments were conducted on a general data test set. When only word roots were used, the performances for the bitext retrieval tasks, s2t and t2s, were 32.03 % and 28.64 %, respectively. These results indicate that although stemming reduces vocabulary diversity, the model is still able to maintain high alignment and retrieval performance in bitext retrieval tasks. This suggests that token alignment models, even in the face of reduced vocabulary diversity, possess excellent generalization and semantic understanding capabilities, effectively enhancing their performance in handling complex language tasks.

5.7. Visualization analysis

The blue dots represent high-resource Chinese medical domain characters, whereas the red dots represent low-resource Mongolian medical domain words. In Fig. 9(a), which presents the results of the mBERT model, while the red dots form noticeable clusters, these clusters are relatively isolated and not closely related to the blue dots. In Fig. 9(b), the DAP model displays a more uniform distribution of points, indicating that the model performed a more extensive dispersion of word embeddings. However, the lack of distinct clustering and strong alignment effects suggests that the model may have limitations in capturing semantic features and performing cross-lingual alignment. In Fig. 9(c), the CLWKD model shows more pronounced clustering, indicating that the model can, to some extent, group semantically similar words together, especially frequently used or contextually similar words. For low-resource languages, this clustering may suggest that the model achieves better semantic alignment for certain vocabularies.

5.8. Performance analysis on high-resource datasets

To validate the model’s performance on high-resource data, we used Chinese-English general data for the first stage of cross-lingual mapping

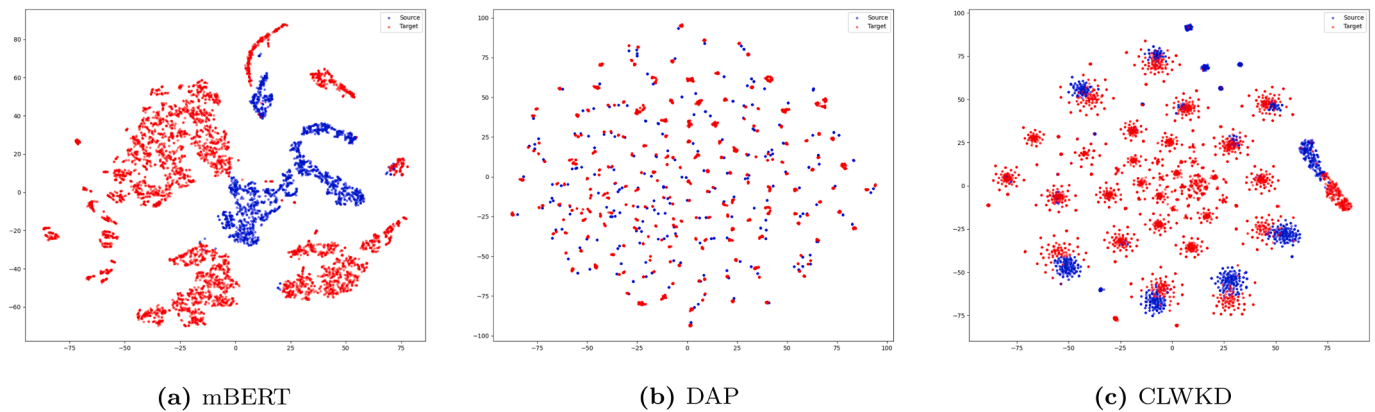


Fig. 9. Visualization of 200 pairs of Mongolian-Chinese medical parallel terms.

Table 12

Cross-lingual sentence retrieval results on the English-Chinese medical domain. The highest values are highlighted in bold, and the second-highest values are underlined.

Model	Teacher	Medical Dataset	
		s2t	t2s
DAP_cls	–	93.40	90.20
AoE	–	95.10	93.60
IMASCL	mBERT	88.22	87.83
IMASCL	mE5	96.20	96.77
CLWKD	MedBERT	73.05	73.24
CLWKD	ERNIE-Health	90.00	89.6
CLWKD	BioBERT	<u>95.72</u>	<u>92.67</u>
CLWKD	MC-BERT	89.42	88.00

training and Chinese-English medical domain data for the second stage of cross-lingual knowledge distillation. In the general domain, we selected 20,000 Chinese-English sentence pairs from the OPUS-Europarl dataset.⁷ In the medical domain, we used 10,000 sentence pairs from the New England Journal of Medicine Chinese-English medical paper dataset⁸. Both datasets were split into training, validation, and test sets at an 8:1:1 ratio.

The experimental results, shown in Table 12, demonstrate that CLWKD is highly competitive compared with recent sentence-level semantic alignment models such as DAP, AoE, and IMASCL in the cross-lingual sentence alignment task. This validates the model's adaptability and generalizability in different resource scenarios, providing strong theoretical and experimental support for low-resource language transfer. As shown in Table 12, in the Chinese-English cross-lingual task, BioBERT, as a teacher model pretrained on English medical data, significantly outperforms other models, indicating its stronger transfer capability in this language direction. This result also reaffirms the conclusion in Section 4.4 that the choice of teacher model has a significant effect on the effectiveness of cross-lingual knowledge distillation.

6. Conclusions

In this paper, a cross-lingual multi-granularity semantic alignment model CLWKD for low-resource languages is proposed, which can be used for both word-level and sentence-level cross-lingual semantic alignment tasks. CLWKD is divided into two stages. In the first stage, multiple

strategies are employed, including agglutinative language morphological decomposition, shared multilingual PLM embedding layers, pretraining the generator, gradient optimization of the mapping matrix, and threshold setting, to learn the multi-granularity cross-lingual mapping matrix for the general domain in an unsupervised and weakly supervised manner. In the second stage, the learned general-domain matrix is applied to cross-lingual knowledge distillation, enhancing domain data and semantic space alignment, thus transferring high-resource domain knowledge to low-resource models while also performing model compression. The experimental results show that CLWKD achieves excellent performance in three cross-lingual tasks (cross-lingual word alignment, dual-text retrieval, and cross-lingual named entity recognition) across Mongolian-Chinese and Korean-Chinese language pairs, covering multiple domains, including medical, legal, and educational. Compared with strong baseline models, CLWKD shows improvements in cross-lingual word alignment tasks. In the Mongolian-Chinese legal and educational domains, the average F1 score increases by 0.14–9.26 %, and in the Korean-Chinese medical and general domains, it improves by 3.23–17.03 %. In dual-text retrieval tasks, CLWKD improves the average F1 score by 0.41–7.12 % in the medical and general domains for both language pairs and by 1.58–24 % in the Mongolian-Chinese legal and educational domains. In the Mongolian-Chinese medical and legal named entity recognition tasks, the average F1 score improves by 1.32–6.57 %. The CLWKD_S model, which is compressed via parameter recycling and bottleneck layer strategies, is also highly competitive compared with the baseline. Additionally, experimental analysis shows that, in agglutinative languages with complex morphology, combining shared multilingual PLM embedding layers and token alignment task training helps improve the model's generalization ability.

Limitations Although our approach is efficient and scalable, we have not yet leveraged high-resource, large-scale medical domain models as teacher models to transfer their knowledge to low-resource models and perform parallel compression owing to computational resource constraints.

CRedit authorship contribution statement

Yurong Wang: Writing – original draft, Software, Methodology; **Min Lin:** Writing – review & editing; **Qitu Hu:** Investigation, Data curation; **Shuangcheng Bai:** Supervision; **Yanling Li:** Supervision; **Longjie Bao:** Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

⁷ <https://object.pouta.csc.fi/OPUS-Europarl/v8/poses/en-zh.txt.zip>

⁸ <https://github.com/boxiangliu/ParaMed>

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